

Applications in shape modelling and registration

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This talk

Two applications

1. Shape modeling and synthesis
2. Image registration

A bit of

- Motivation
- Inspiration
- Experiences
- Challenges (and some solutions)

Shape

Shape representations



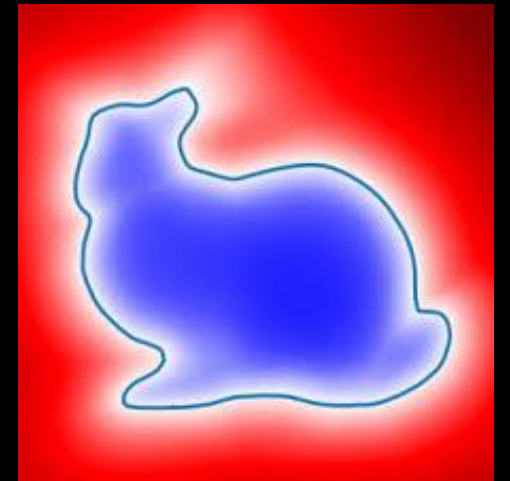
Voxels



Points



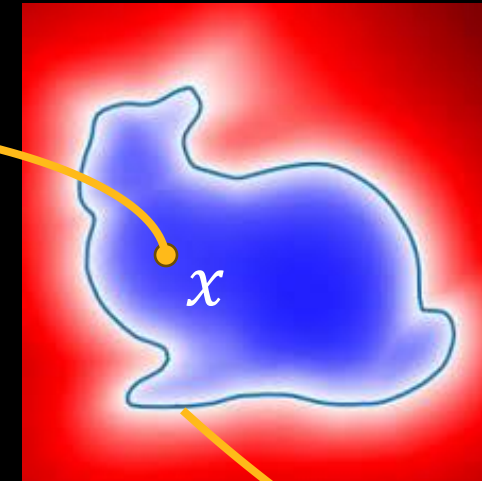
Meshes



Functions

Signed distance function

$$SDF_{\mathcal{S}}(x) = \begin{cases} -\|x - \mathcal{S}\| & \text{if } x \text{ inside } \mathcal{S} \\ 0 & \text{if } x \text{ on } \mathcal{S} \\ \|x - \mathcal{S}\| & \text{if } x \text{ outside } \mathcal{S} \end{cases}$$



Zero level set

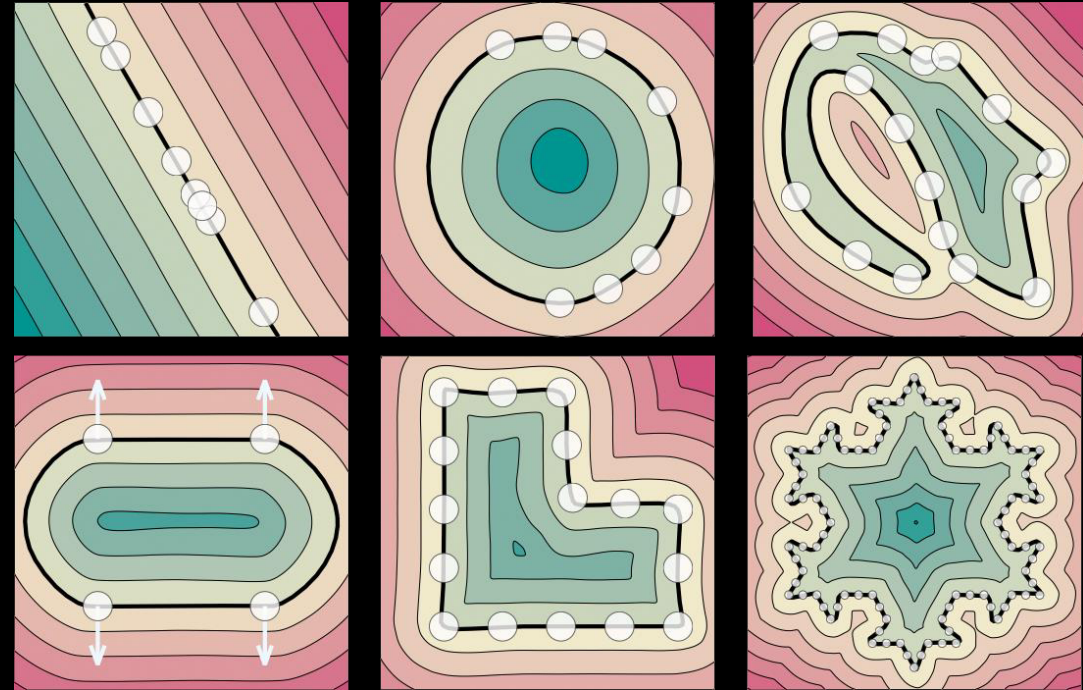
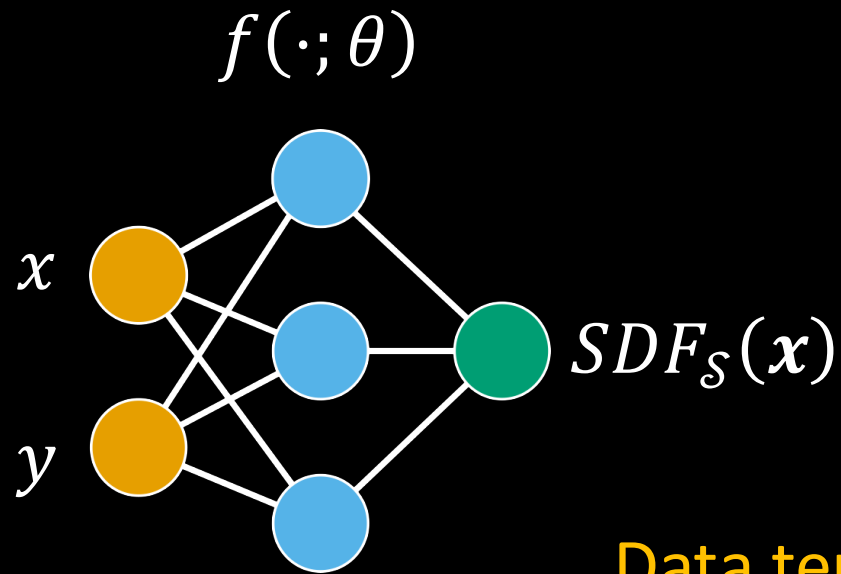
□ Represent SDF as neural field

Fully supervised fitting



Semi-supervised fitting

Fit to N points

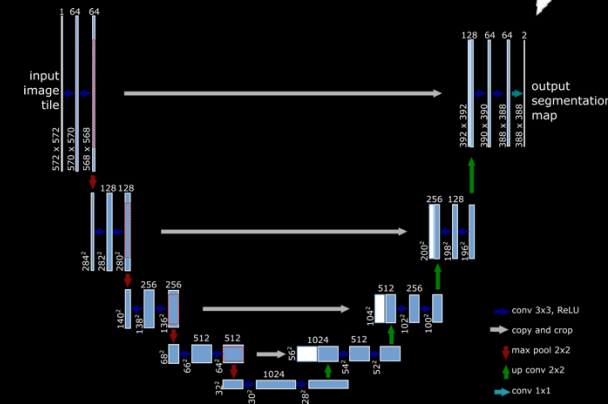


Data term

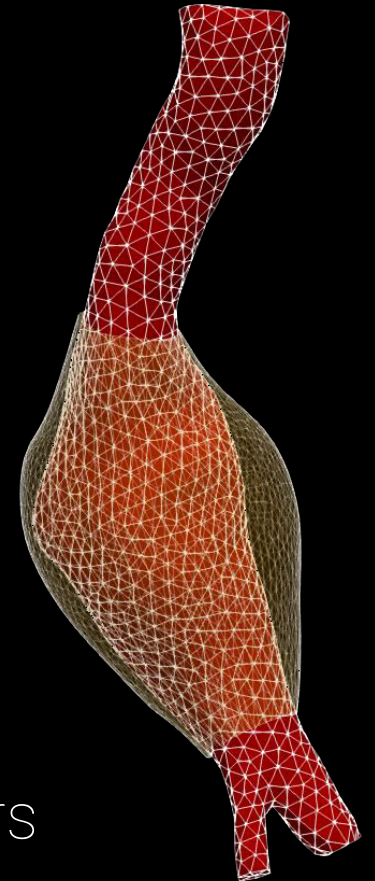
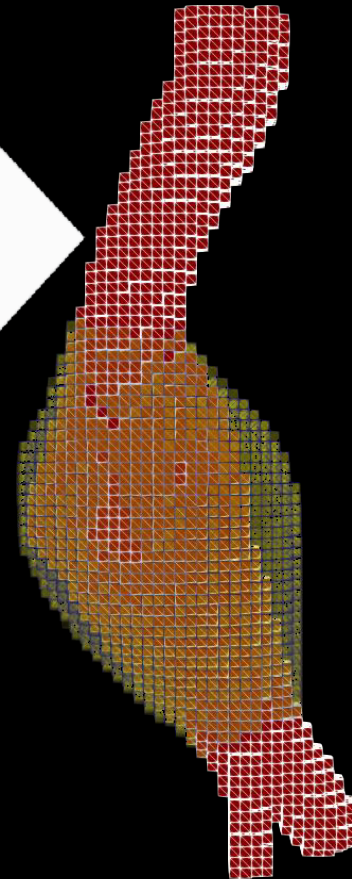
$$\ell(\theta) = \frac{1}{N} \sum_{1 \leq i \leq N} (|f(\mathbf{x}_i; \theta)|) + \lambda \mathbb{E}_x (\|\nabla_x f(\mathbf{x}; \theta)\| - 1)^2$$

Eikonal term

Application: artery segmentation



U-Net



- ✓ Priors
- ✓ Symmetries

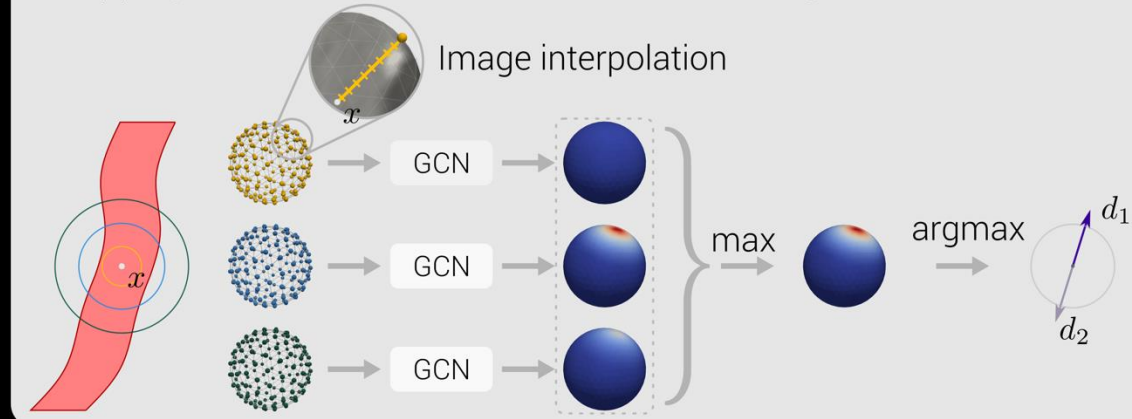
Segmenting with symmetries



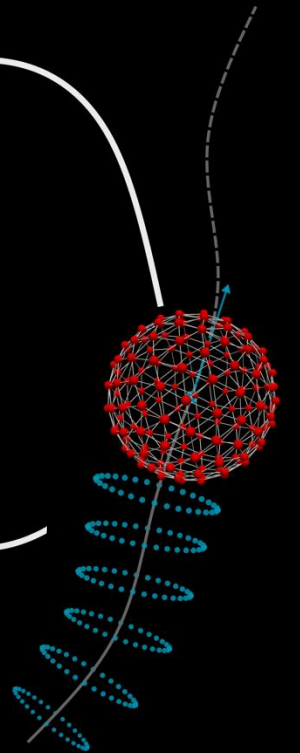
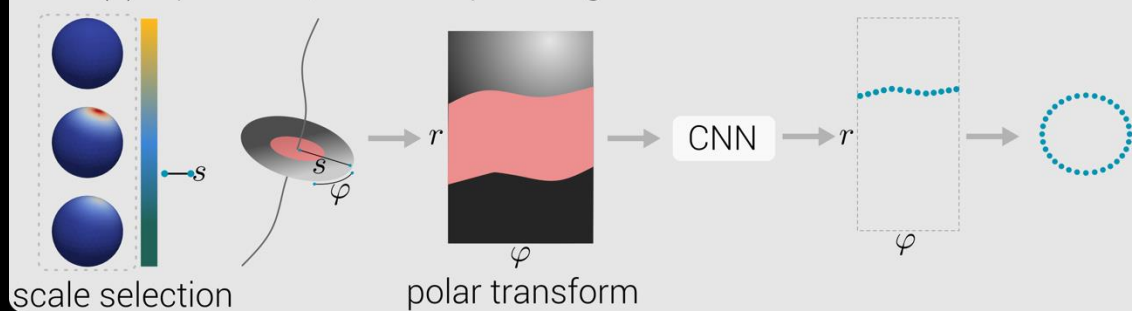
Dieuwertje Alblas



1. $SO(3)$ equivariant, scale invariant centerline tracking



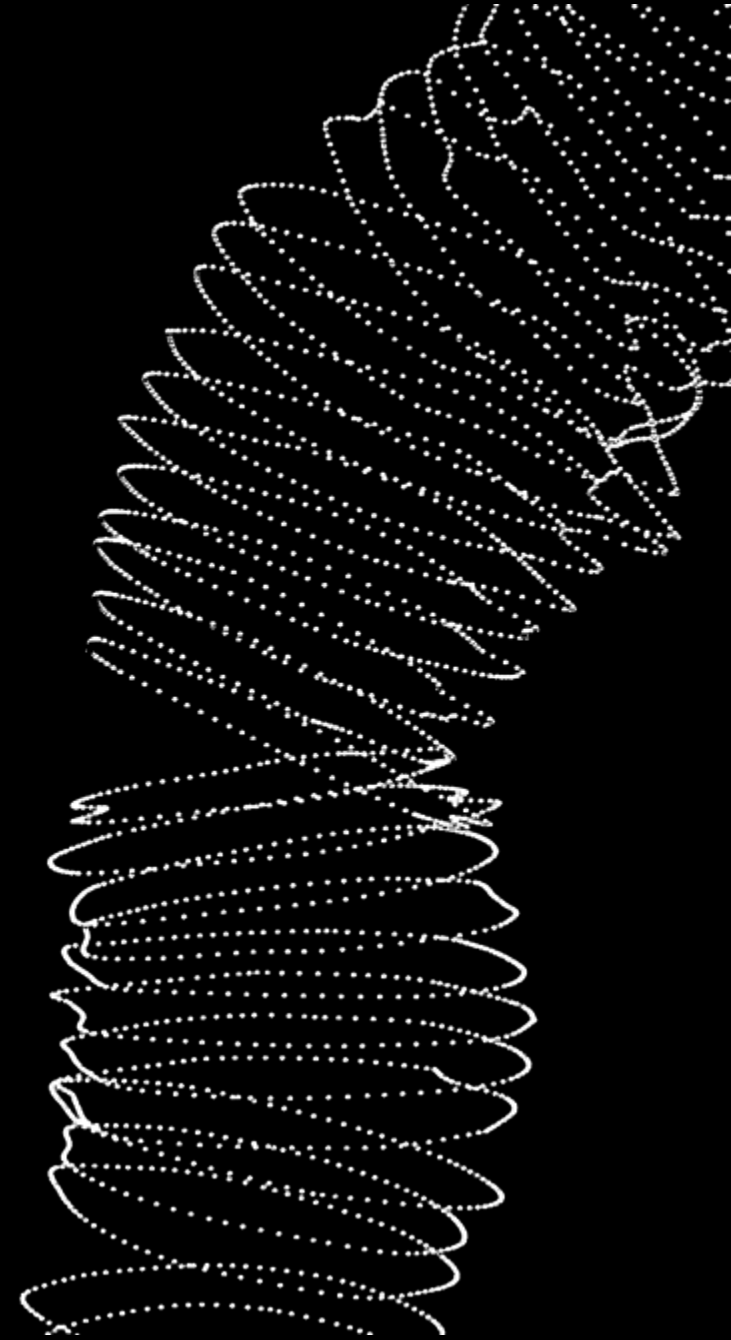
2. $SO(2)$ equivariant, scale adaptive segmentation



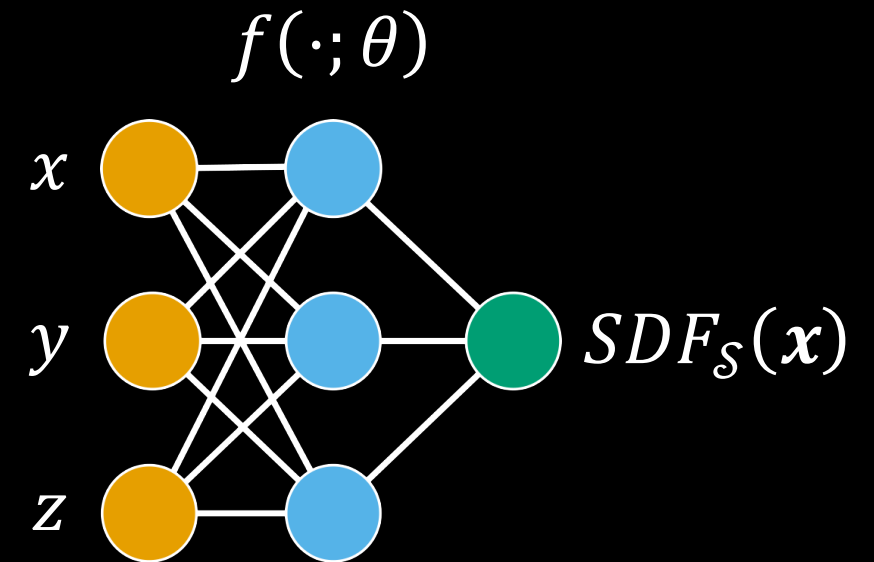
From contours to surface

- Voxelization is undesirable
 - Limited resolution
 - Loss of accuracy
- Meshing is challenging
 - Intersecting contours
 - Unequal spacing

□ Fit neural SDF to point cloud

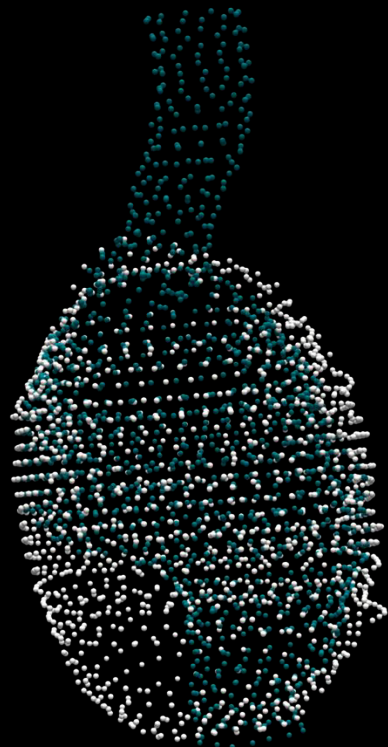


Signed distance functions in 3D

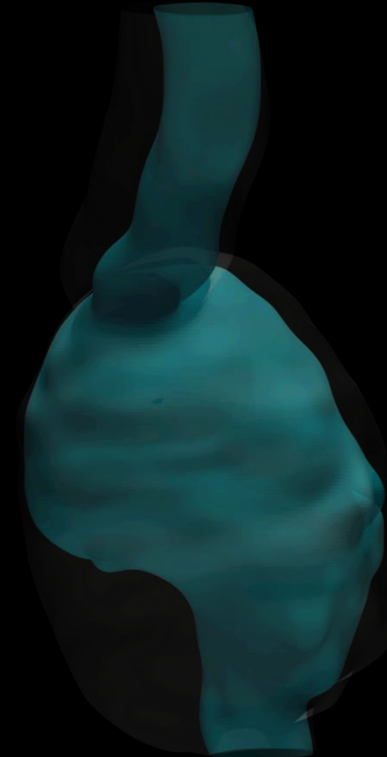
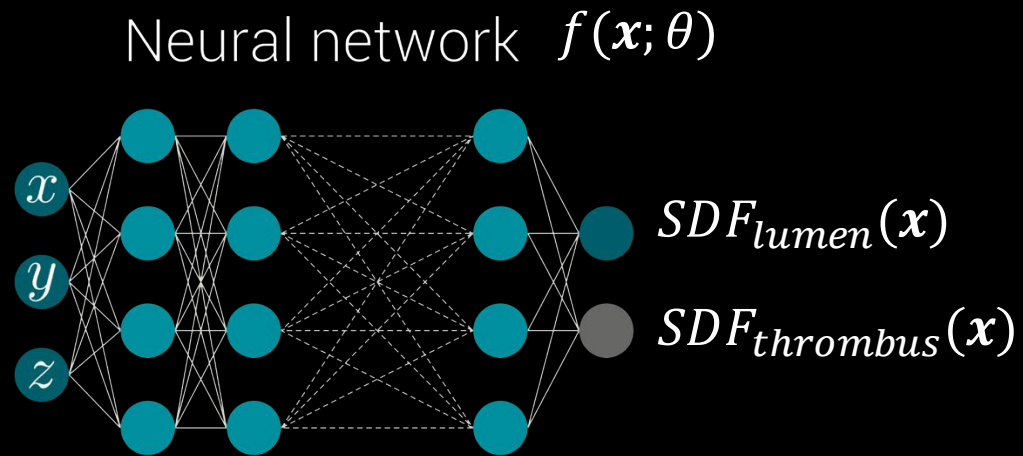


$$\ell(\theta) = \frac{1}{N} \sum_{1 \leq i \leq N} (|f(\mathbf{x}_i; \theta)|) + \lambda \mathbb{E}_{\mathbf{x}} (\|\nabla_{\mathbf{x}} f(\mathbf{x}; \theta)\| - 1)^2$$

Joint fitting of multiple surfaces

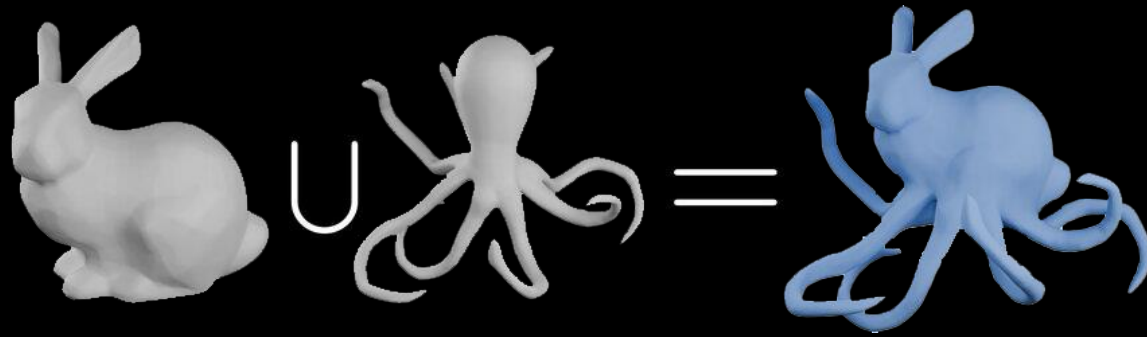


Pointcloud



Surface reconstruction

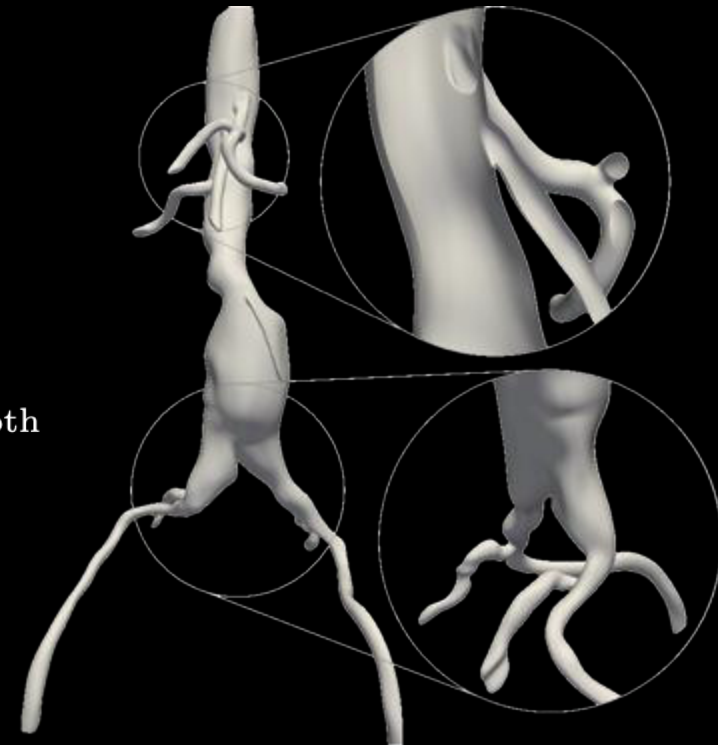
Smooth blending of SDFs



$$\mathcal{M}_1 \cup \mathcal{M}_2 = \min(SDF_{\mathcal{M}_1}(\mathbf{x}), SDF_{\mathcal{M}_2}(\mathbf{x}))$$

$$(\mathcal{M}_1 \cup \mathcal{M}_2)_{\text{smooth}} = \min(SDF_{\mathcal{M}_1}(\mathbf{x}), SDF_{\mathcal{M}_2}(\mathbf{x})) - \gamma_{\text{smooth}}$$

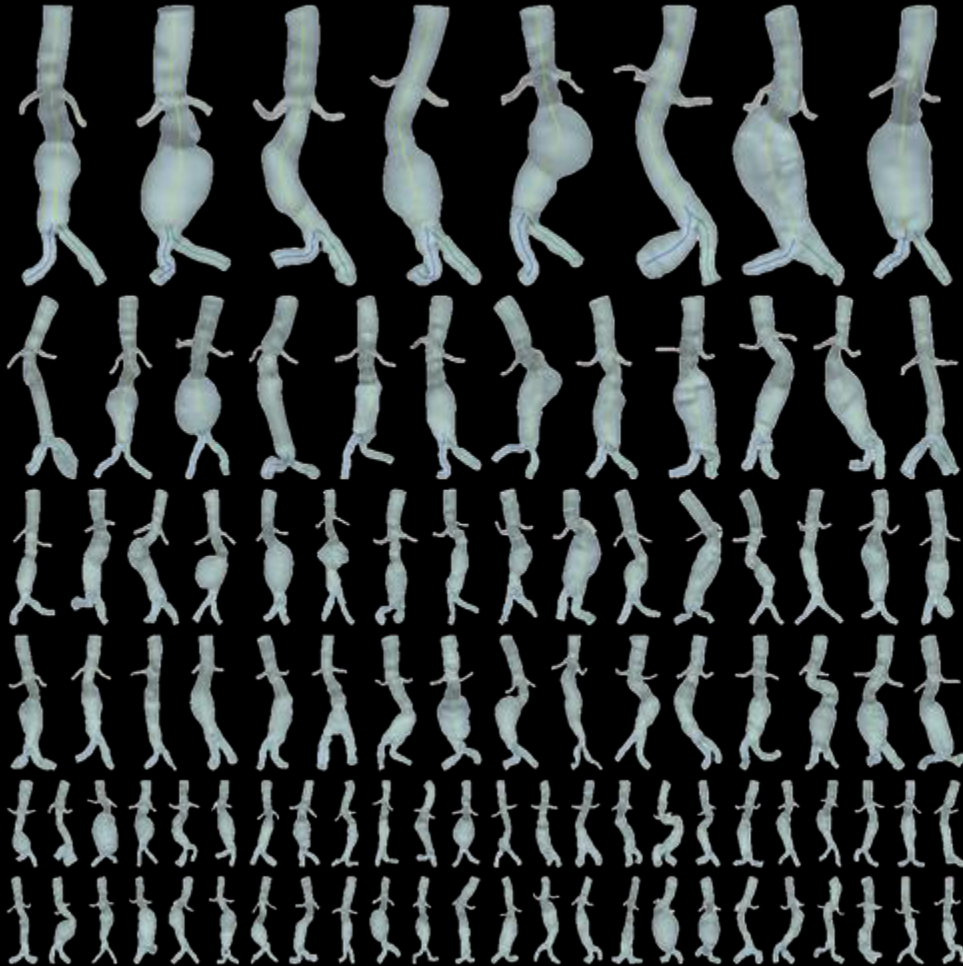
$$\gamma_{\text{smooth}} = \frac{1}{4k} \cdot \max(k - |SDF_{\mathcal{M}_1}(\mathbf{x}) - SDF_{\mathcal{M}_2}(\mathbf{x})|, 0)^2,$$



Fully automatic pipeline



Patryk Rygiel



Challenges

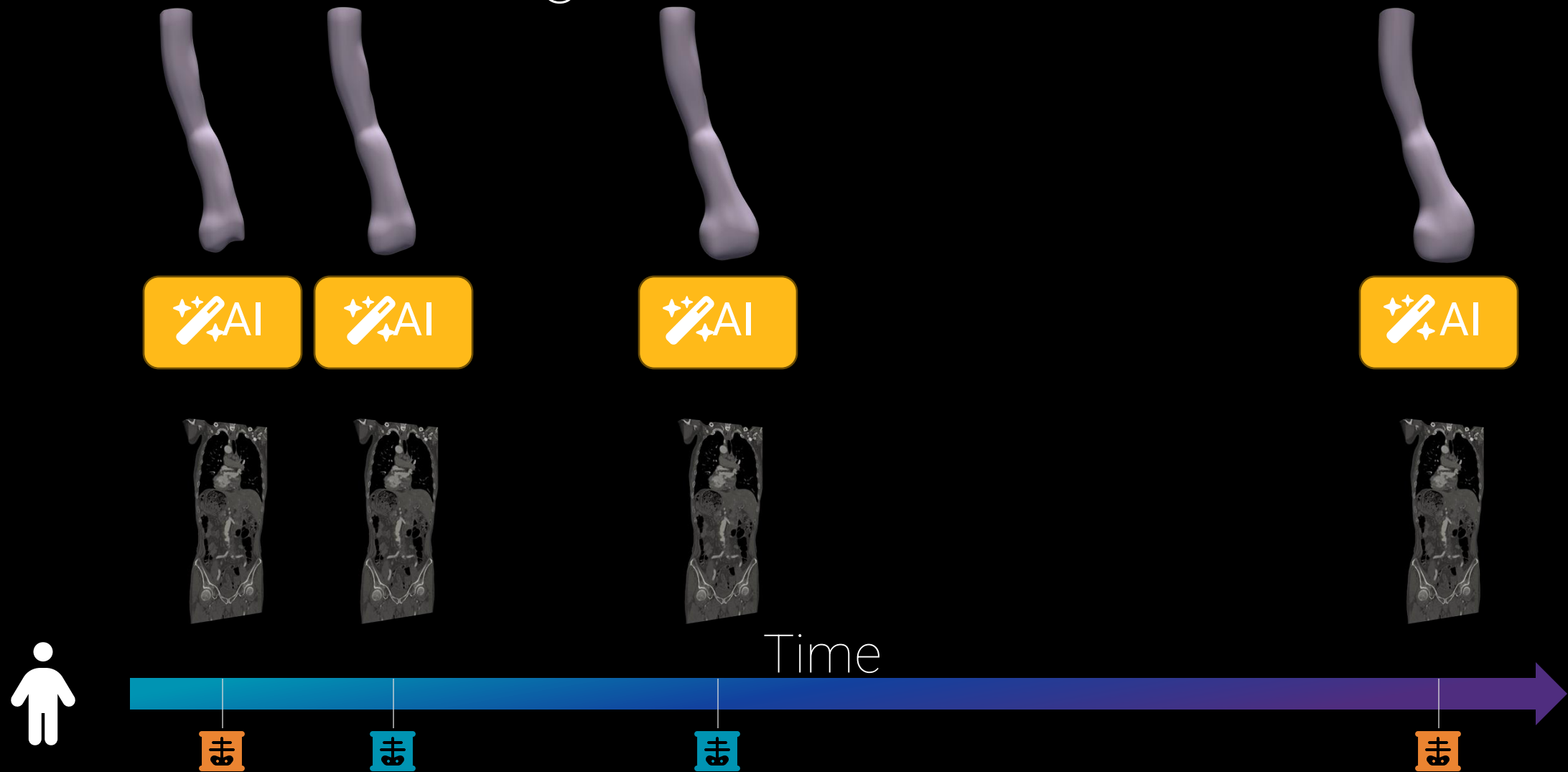
- Optimization sensitive to initialization
- Structures vary in smoothness

Solutions

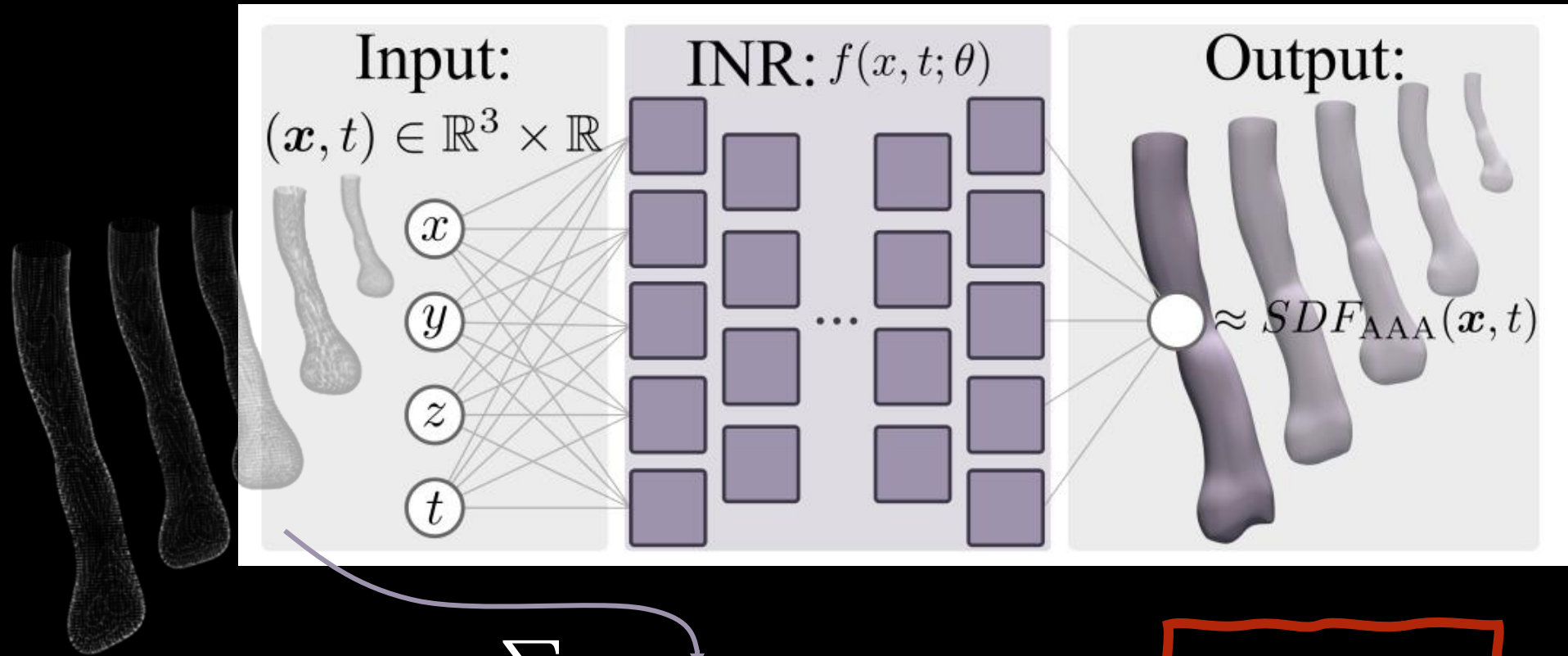
- Regularization beyond Eikonal¹
- Posthoc quality assessment
 - Smoothness
 - Topology: Euler characteristic $\chi = V - E + F$

¹ Ma, B., Han, Z., Liu, Y. S., & Zwicker, M. (2021, July). Neural-Pull: Learning Signed Distance Function from Point clouds by Learning to Pull Space onto Surface. In ICML (pp.7246-7257). PMLR.

Growth modeling

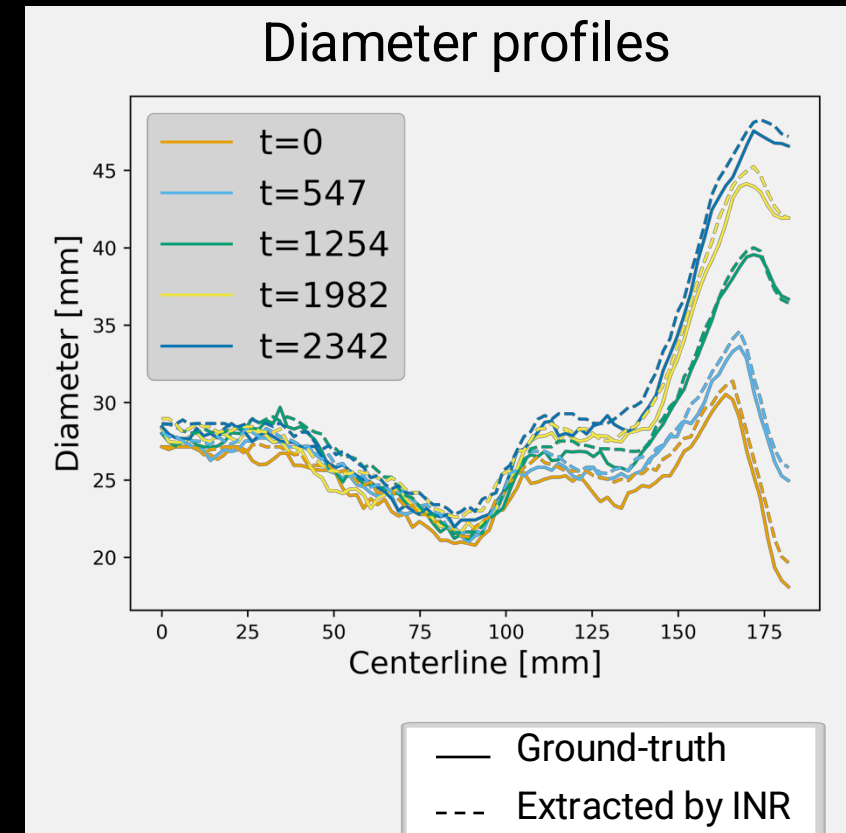
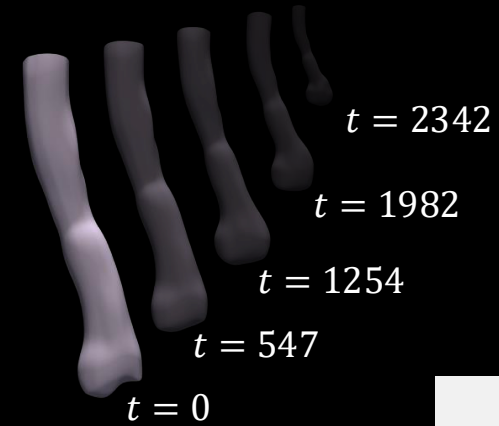
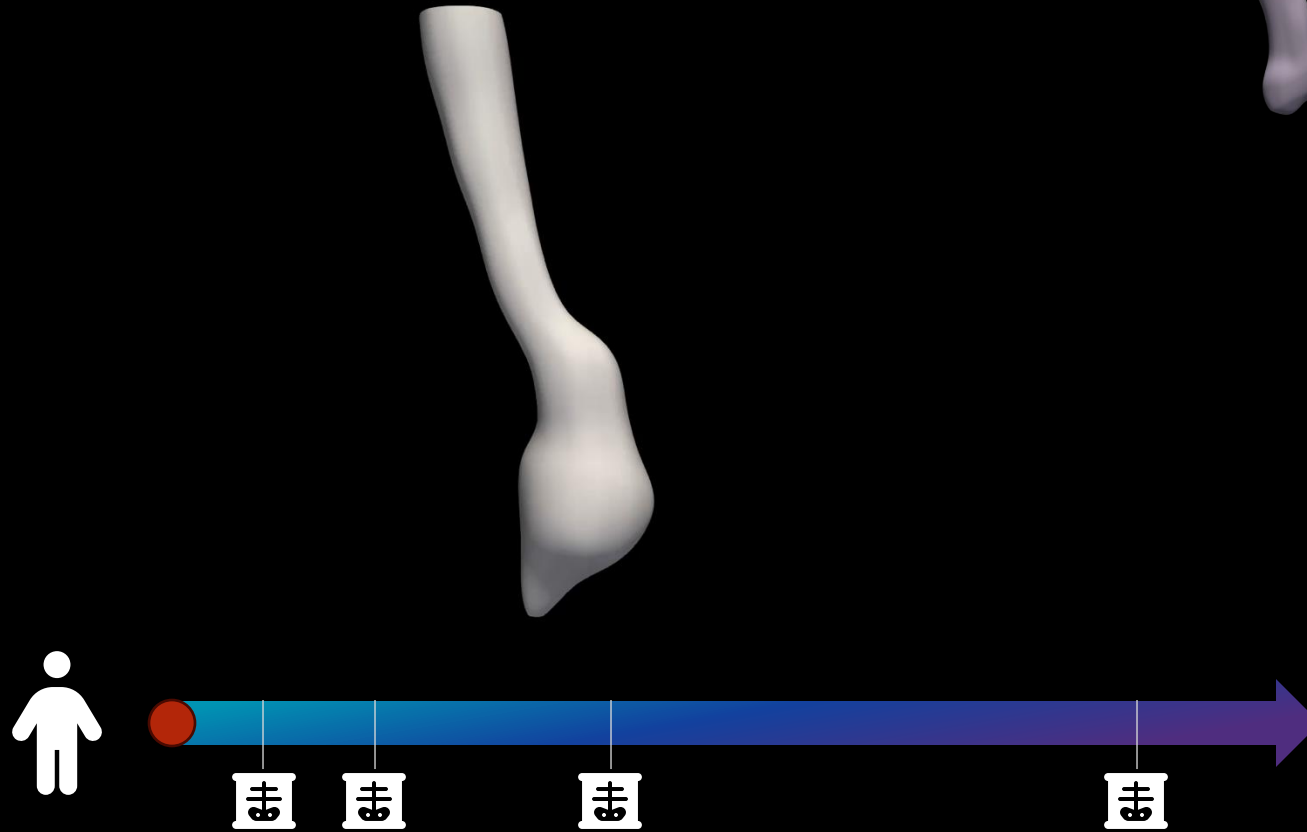


Modeling growth

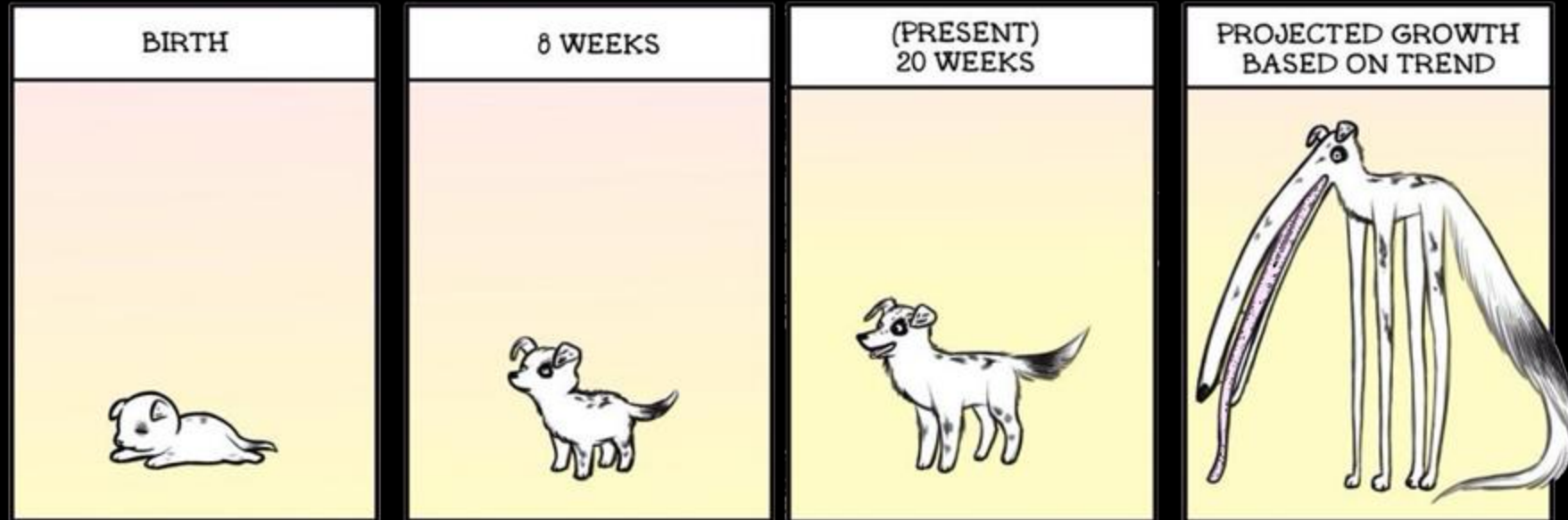


$$\mathcal{L}(\theta) = \sum_{1 \leq j \leq J} \mathcal{L}_{data_j}(\theta) + \lambda_1 \mathcal{L}_{Eikonal}(\theta) + \lambda_2 \mathcal{L}_{temporal}(\theta)$$

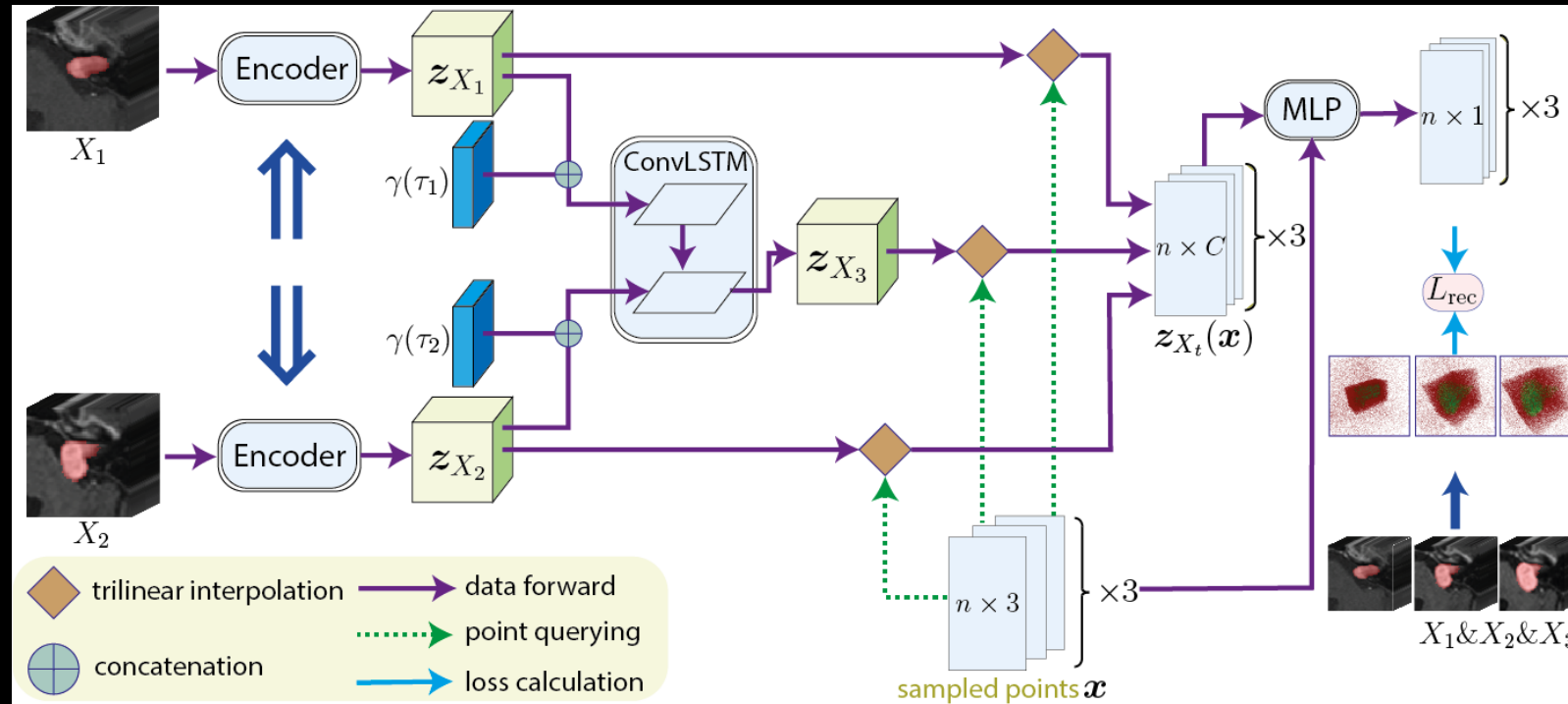
Modeling growth



Extrapolation $\neq$ interpolation



Growth prediction



T-AM-281

Vestibular schwannoma growth prediction from longitudinal MRI by time-conditioned neural fields
Yunjie Chen, Jelmer M. Wolterink, Olaf M. Neve, Stephan R. Romeijn, Berit M. Verbist, Erik F. Hensen, Qian Tao, Marius Staring

Cell shape synthesis

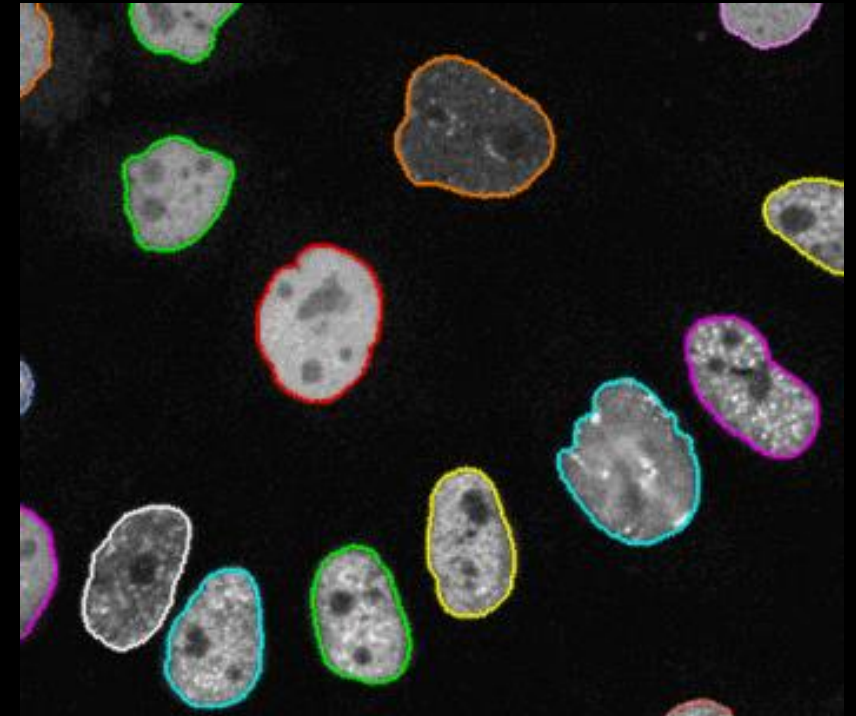
Goals

1. Synthesize new 3D+t living cell shapes
2. Synthesize corresponding image

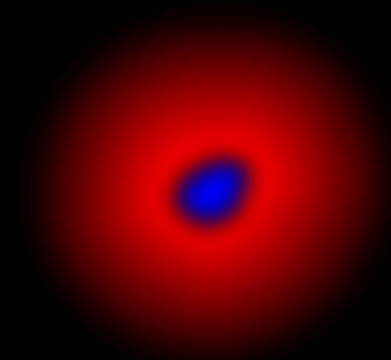
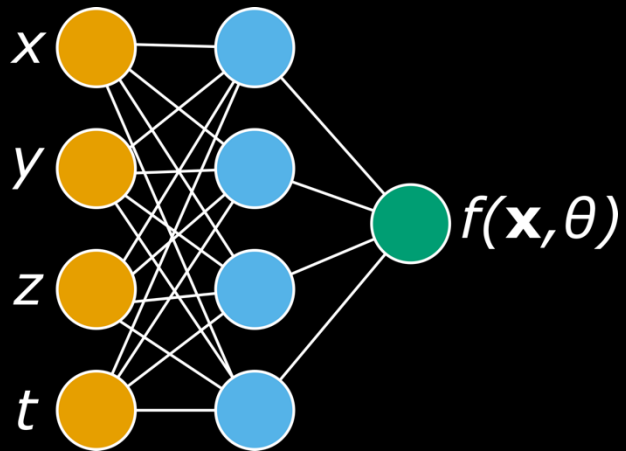
Existing models

- Parametric
 - Ellipses, elliptical Fourier descriptors, spherical harmonics, meshes, active contours, SSMs, ...
- Deep learning-based voxel volumes

□ Represent living cell as neural SDF



Modeling a single cell



Zero level set
→

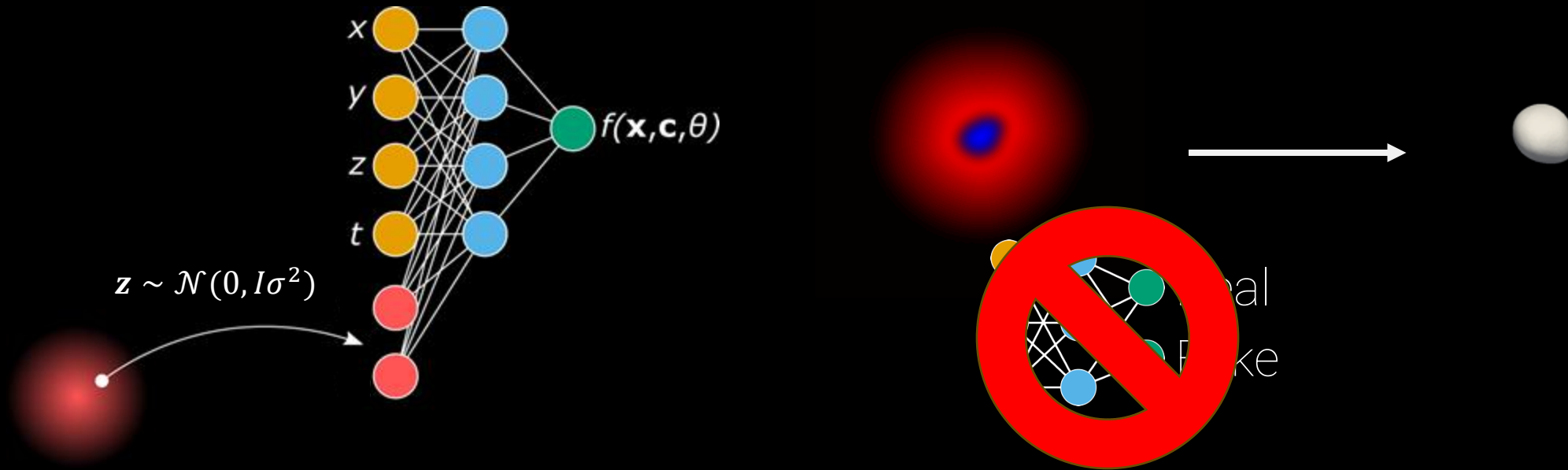


Modeling multiple cells

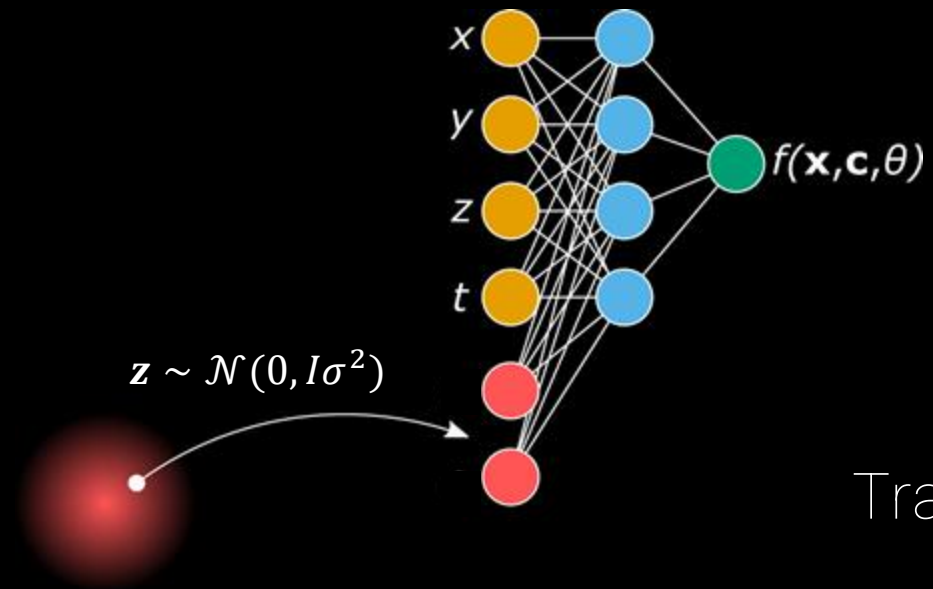
- Generative model
- INR as generator/decoder, but no discriminator/encoder ...
- Auto-decoder!



Functa/meta-learning



Auto-decoder

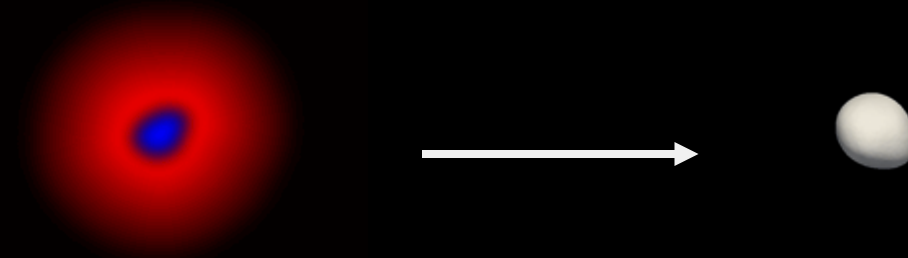


Training

$$\arg \min_{\theta, \{z_i\}_{i=1}^N} \sum_{i=1}^N \left(\sum_{j=1}^K \mathcal{L}(f_{\theta}(z_i, \mathbf{x}_j), s_j) + \frac{1}{\sigma^2} \|z_i\|_2^2 \right)$$

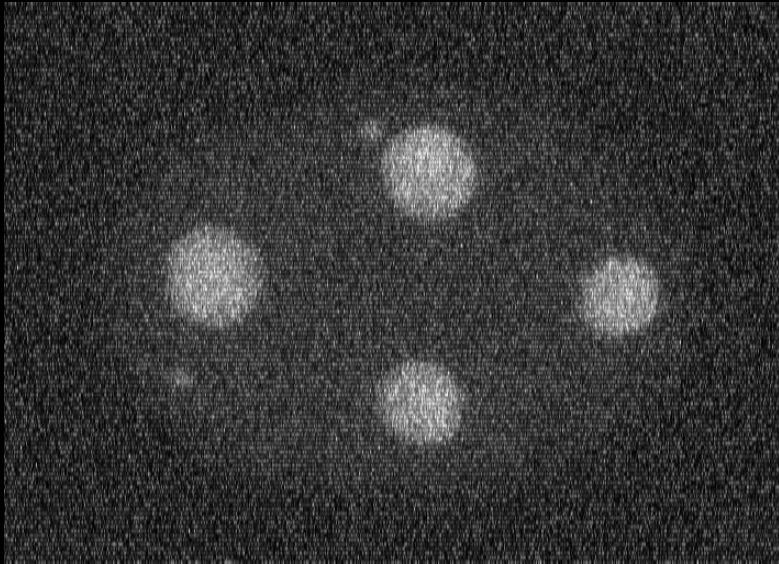
Inference

$$\hat{z} = \arg \min_z \sum_{(\mathbf{x}_j, s_j) \in X} \mathcal{L}(f_{\theta}(z, \mathbf{x}_j), s_j) + \frac{1}{\sigma^2} \|z\|_2^2$$



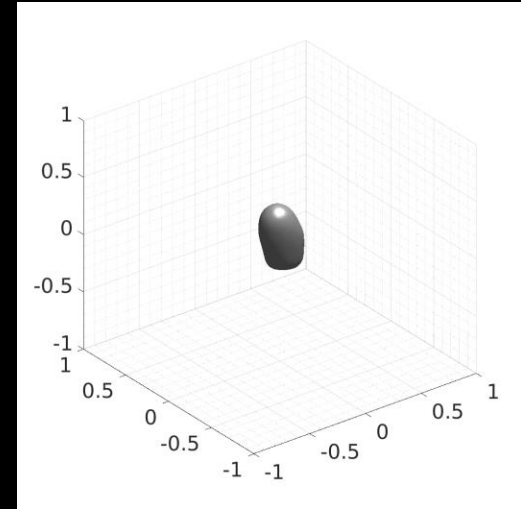
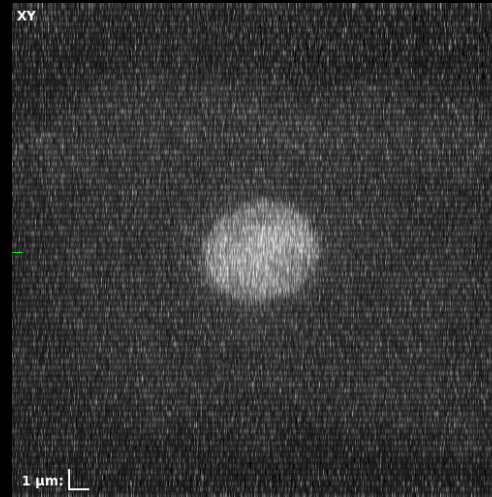
C. elegans

Real



Synthetic

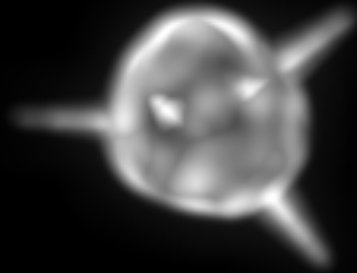
pix2pix



Time:

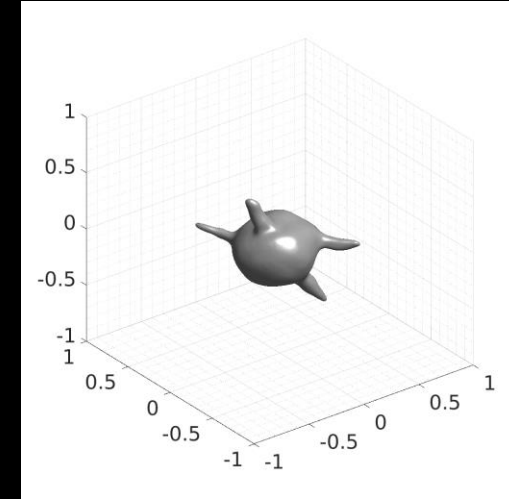
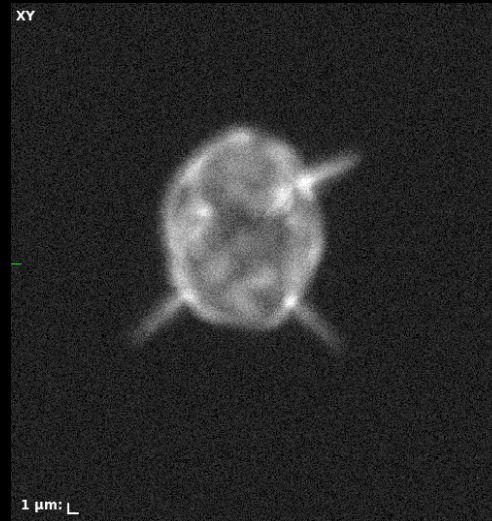
A549 Human Lung Carcinoma

Real



Synthetic

pix2pix



Time: _____

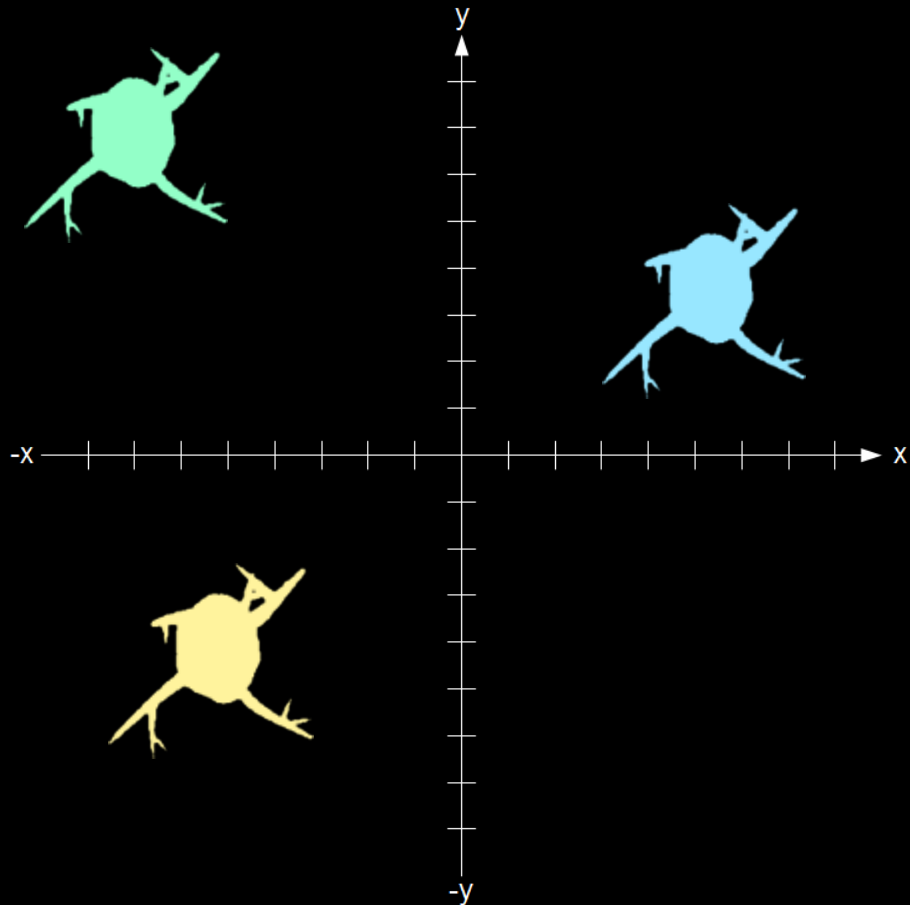
What is shape?

'The shape of a set of points is all the geometrical information that is invariant to translations, rotations, and size changes'

Problem: INRs assume common coordinate system!

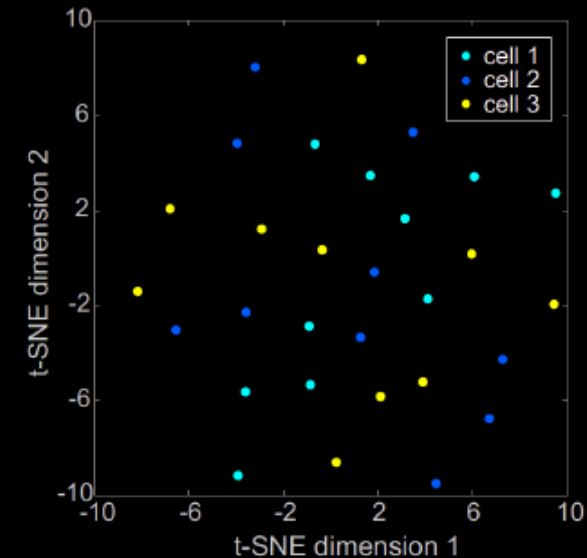
Focus on shape

Translation: shift all cells to center of coordinate space



Rotation experiment

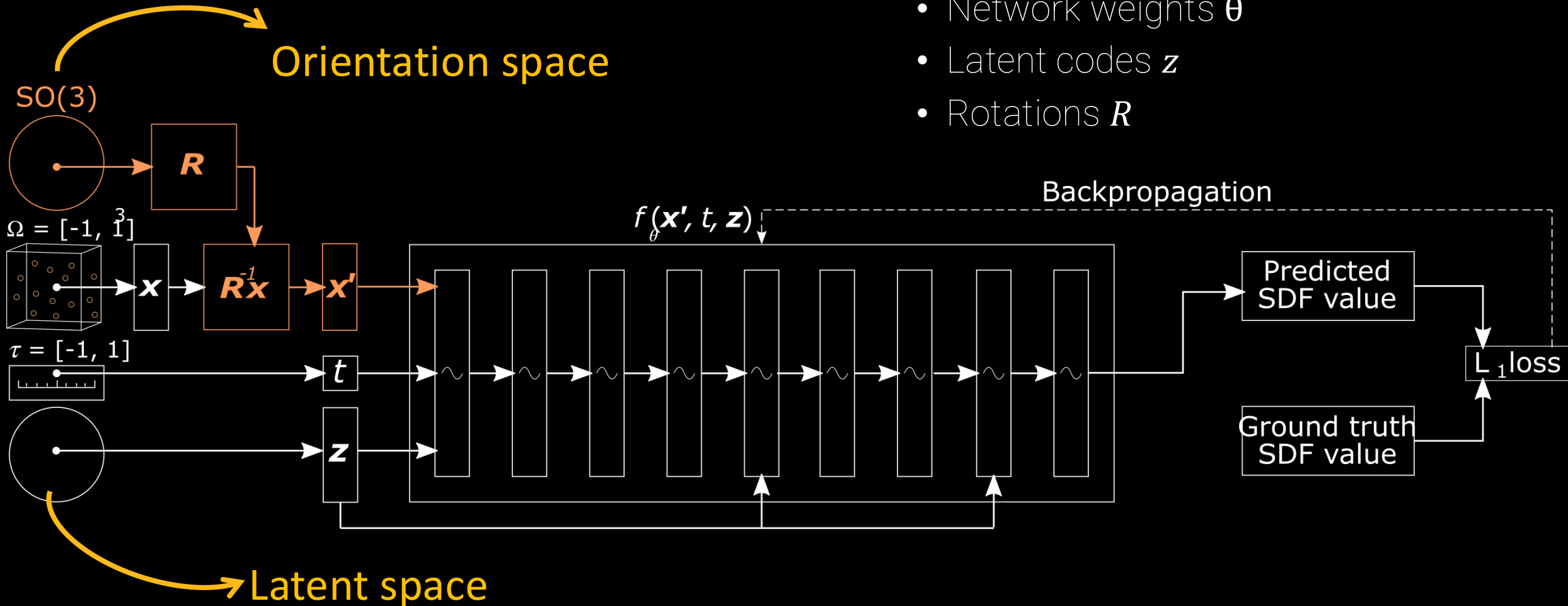
- Optimize/encode latent code
- Latent code = shape + rotation



Rotation disentanglement

Jointly optimize

- Network weights θ
- Latent codes $\mathbf{z}$
- Rotations $\mathbf{R}$

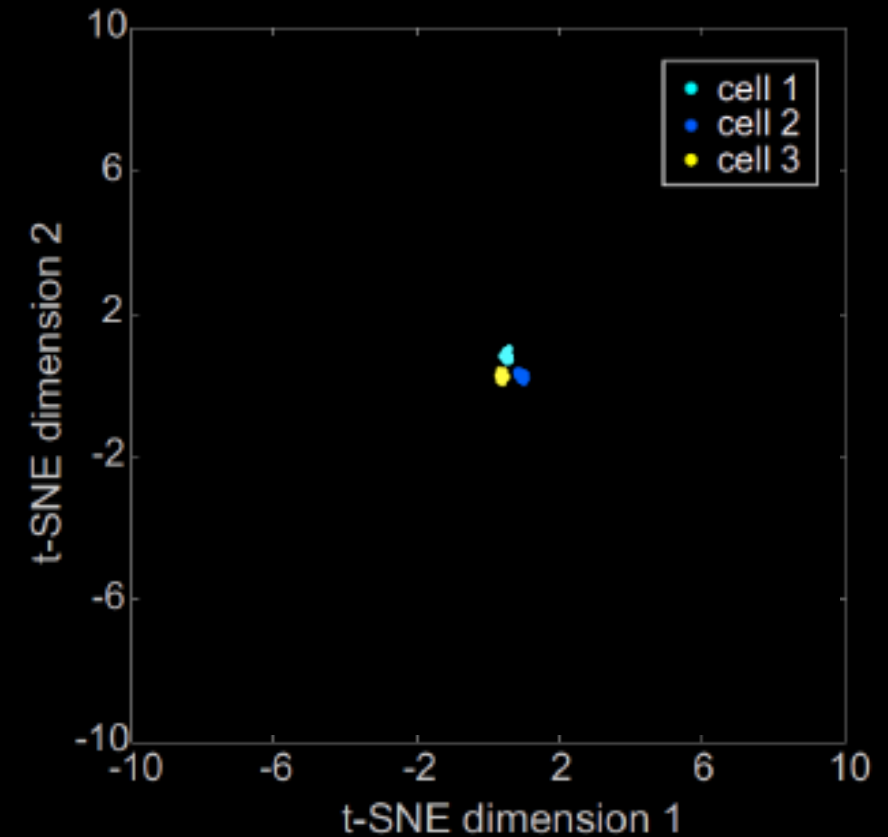


Result



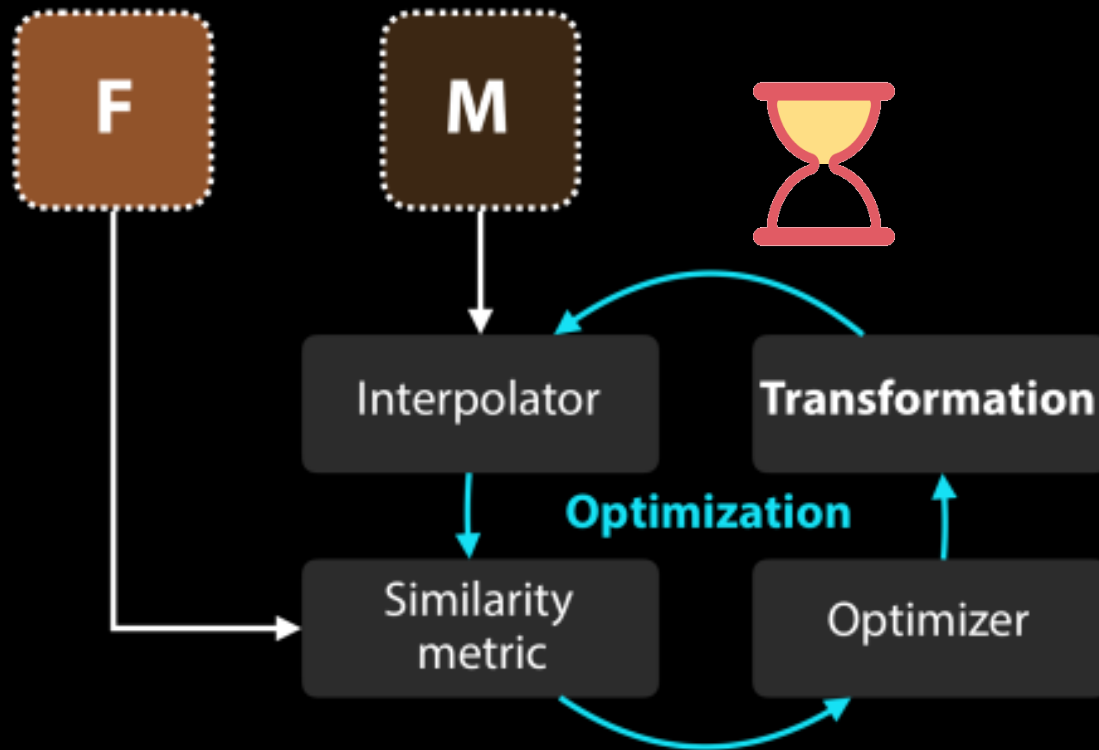
David Wiesner

- Same setting
- New embeddings
- Latent code == shape



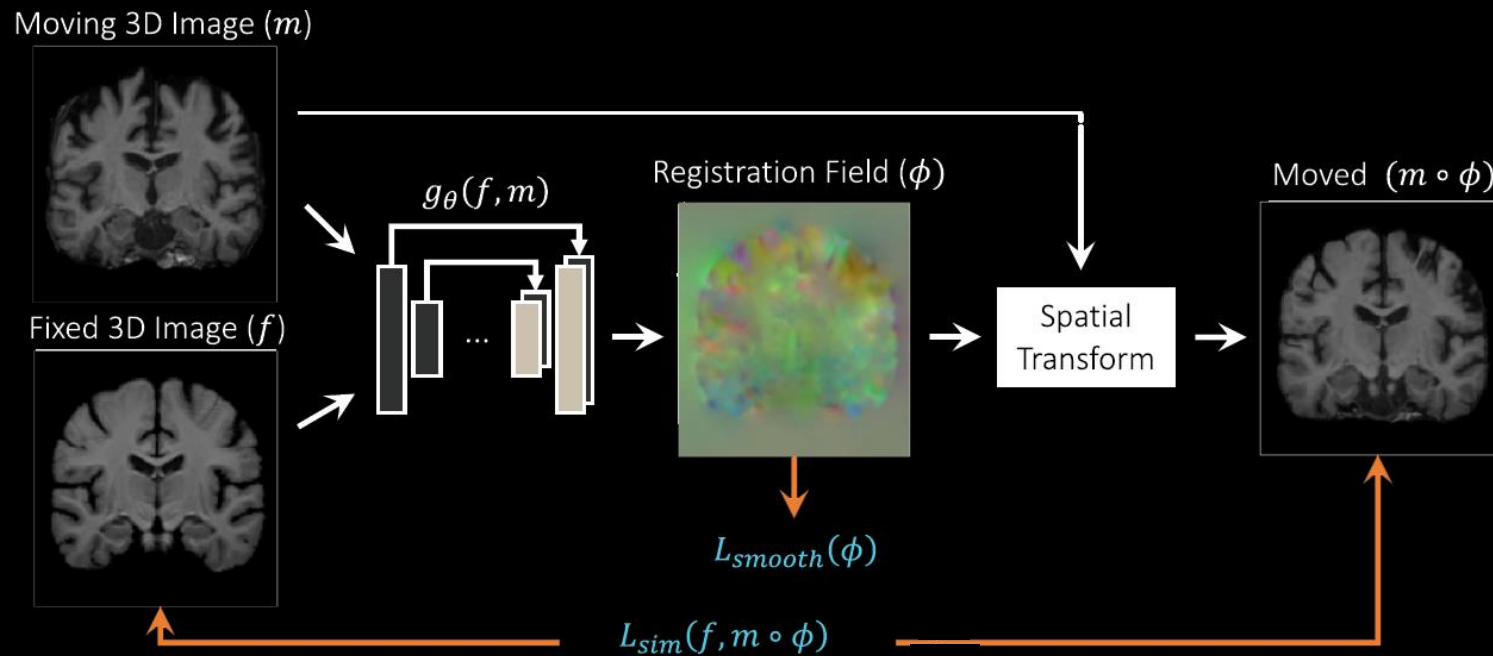
Registration

Image registration



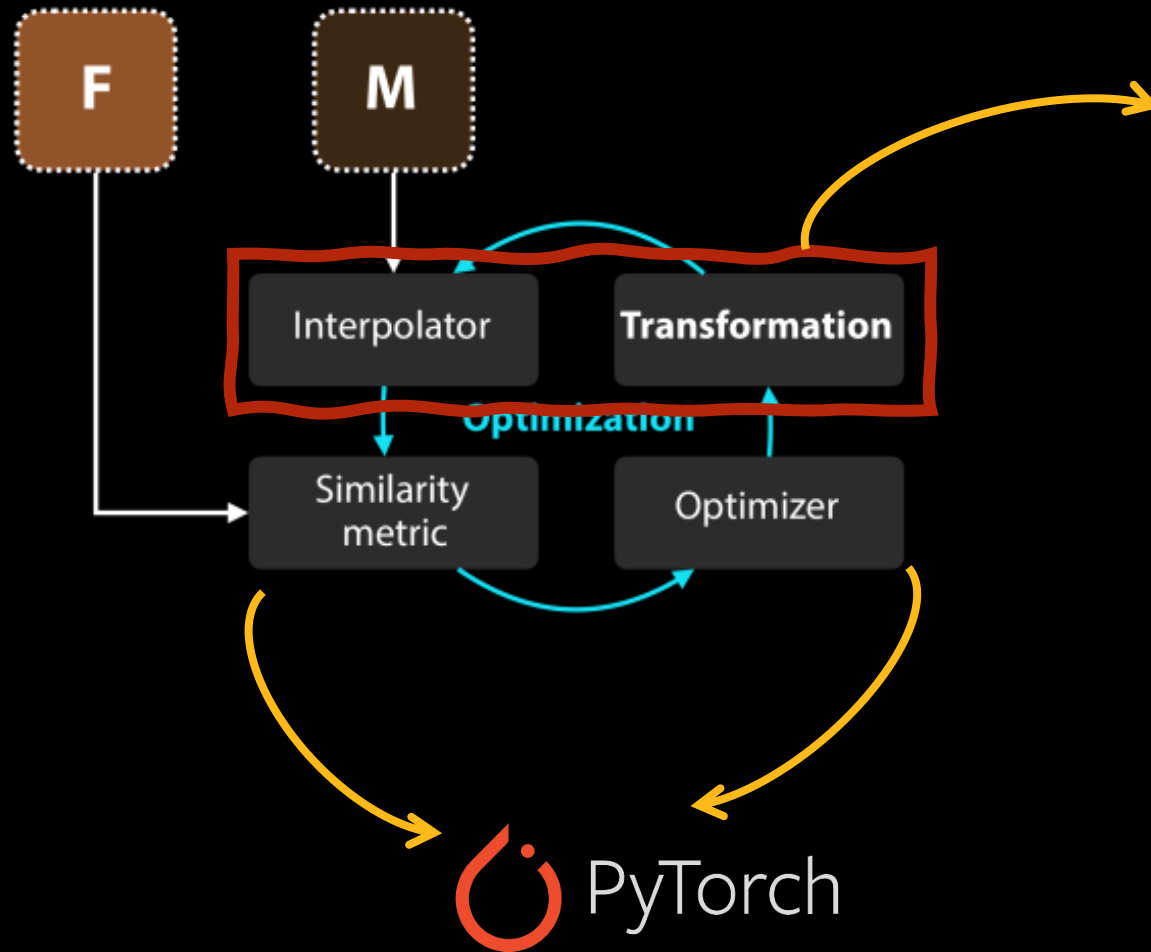
🔍 Deep learning!

Deep learning registration



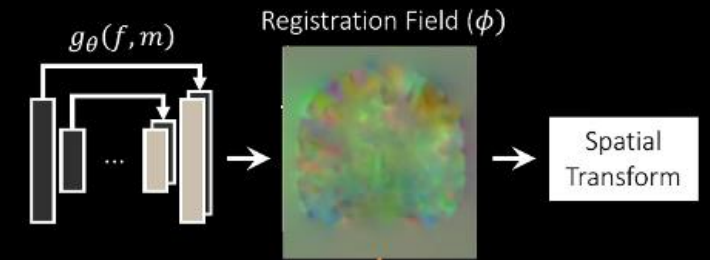
- ✓ Generalization
- ❖ Need for training data
- ❖ Performance

Iterative optimization revisited

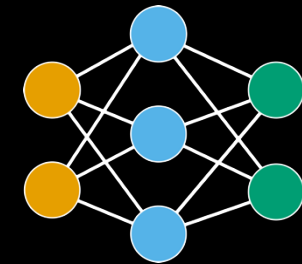


One-shot image registration *aka* instance optimization

Option 1



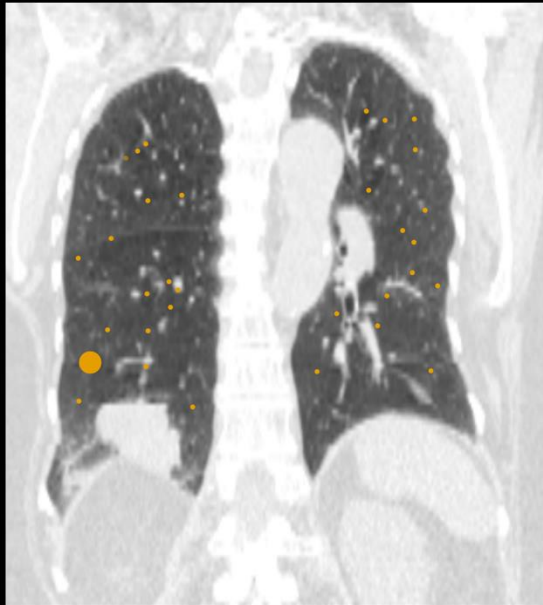
Option 2



□ Transformation as INR

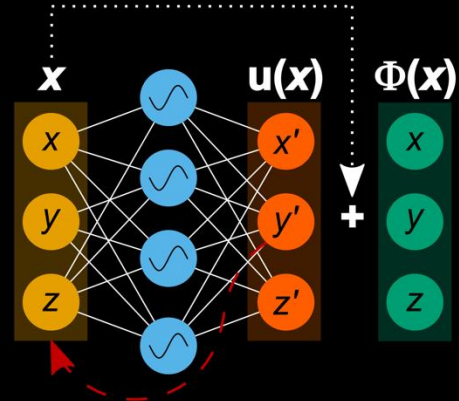
Deformation as INR

Fixed image F



$$F: \Omega_F \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

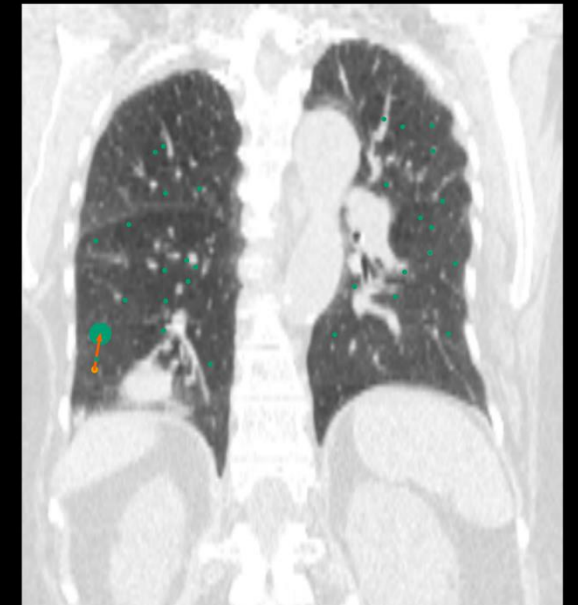
Transformation Φ



Objective

$$\hat{\Phi} = \operatorname{argmin}_{\Phi} \mathcal{L}_{data}(M \circ \Phi, F) + \alpha \mathcal{L}_{reg}(\Phi)$$

Moving image M



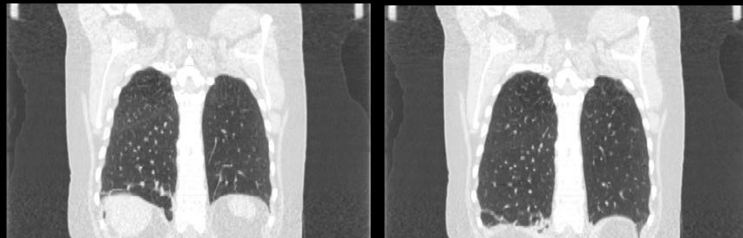
$$M: \Omega_M \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

ReLU: Activation functions matter

Deformed image



Deformation magnitude



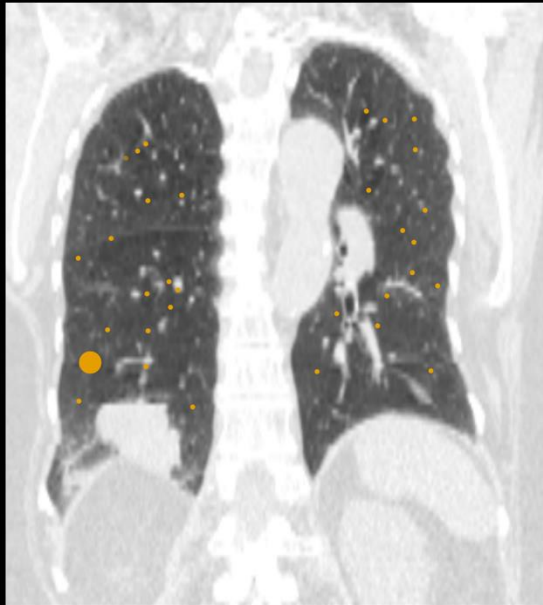
M

F

TRE = 1.43 mm

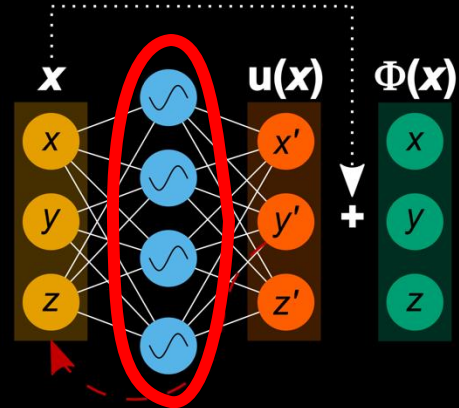
Deformation as INR

Fixed image F



$$F: \Omega_F \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

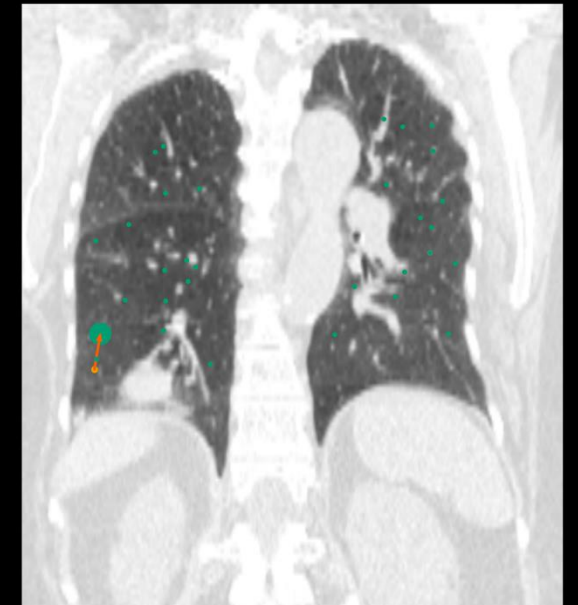
Transformation Φ



Objective

$$\hat{\Phi} = \operatorname{argmin}_{\Phi} \mathcal{L}_{data}(M \circ \Phi, F) + \alpha \mathcal{L}_{reg}(\Phi)$$

Moving image M



$$M: \Omega_M \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

SIREN

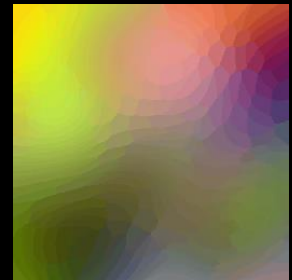
The SIREN model contains a parameter ω to adjust frequency

$$y = \sin(\omega Wx + b)$$

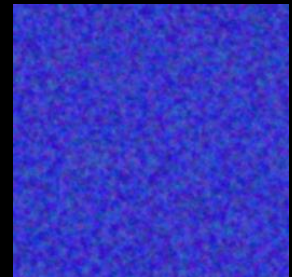
Intuition

- Low $\omega \rightarrow$ low-frequency signal
- High $\omega \rightarrow$ high-frequency signal

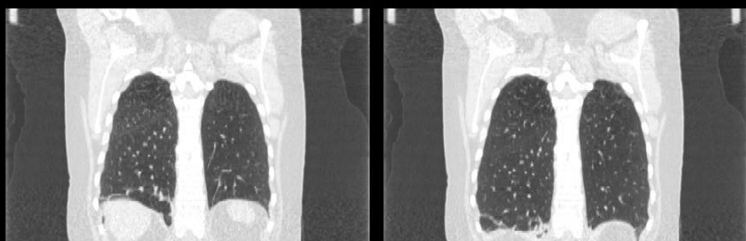
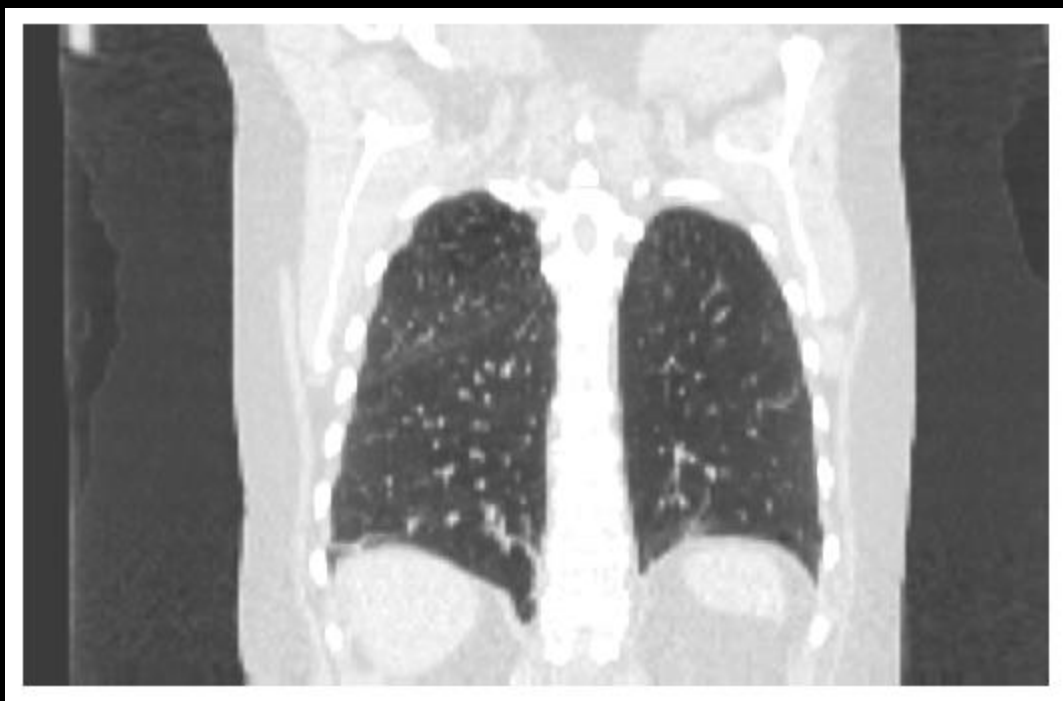
$$\omega = 4$$



$$\omega = 128$$



SIREN: $\omega = 8$



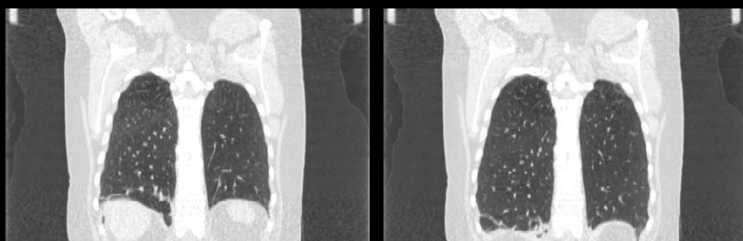
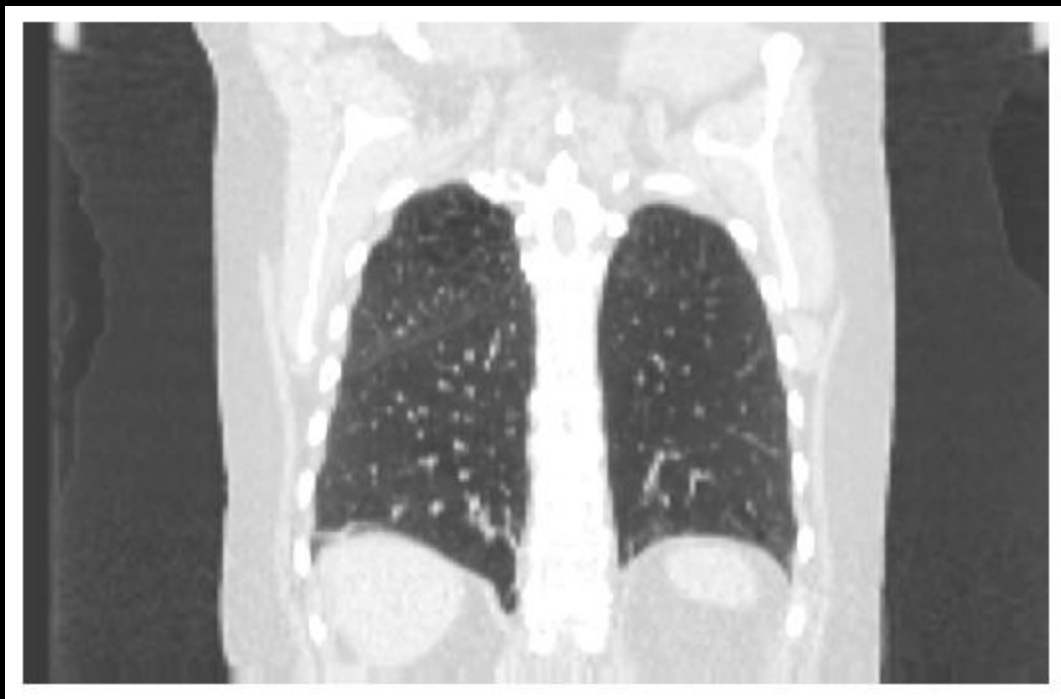
M

F



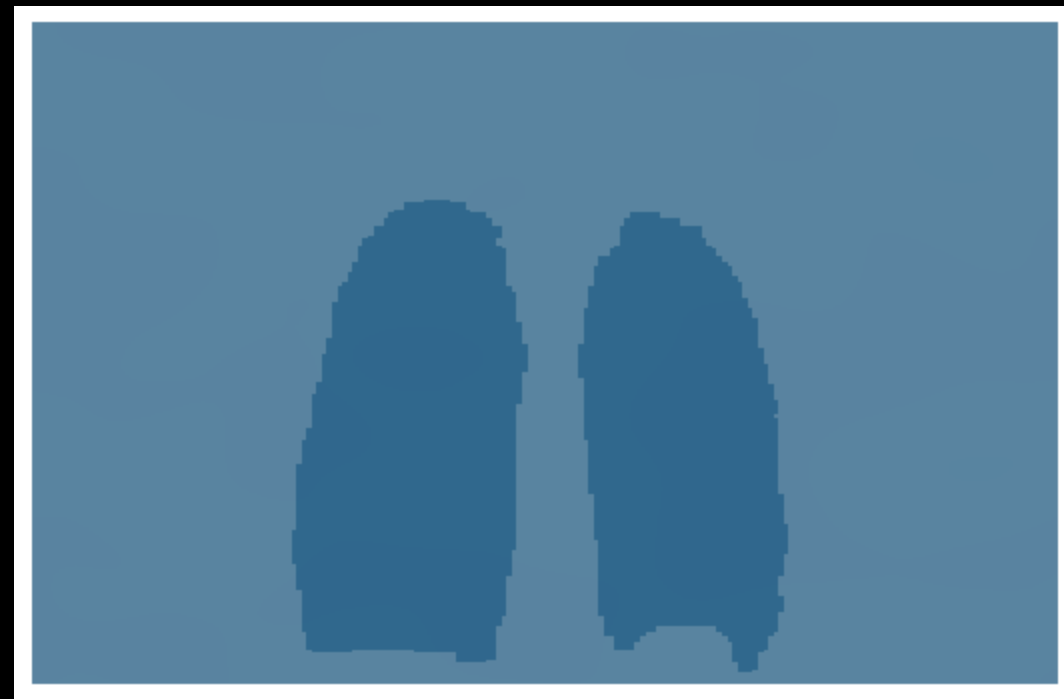
TRE = 1.31 mm

SIREN: $\omega = 32$



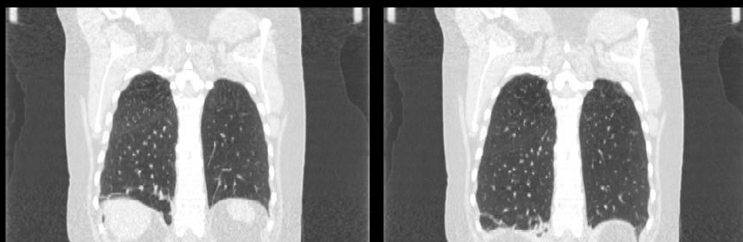
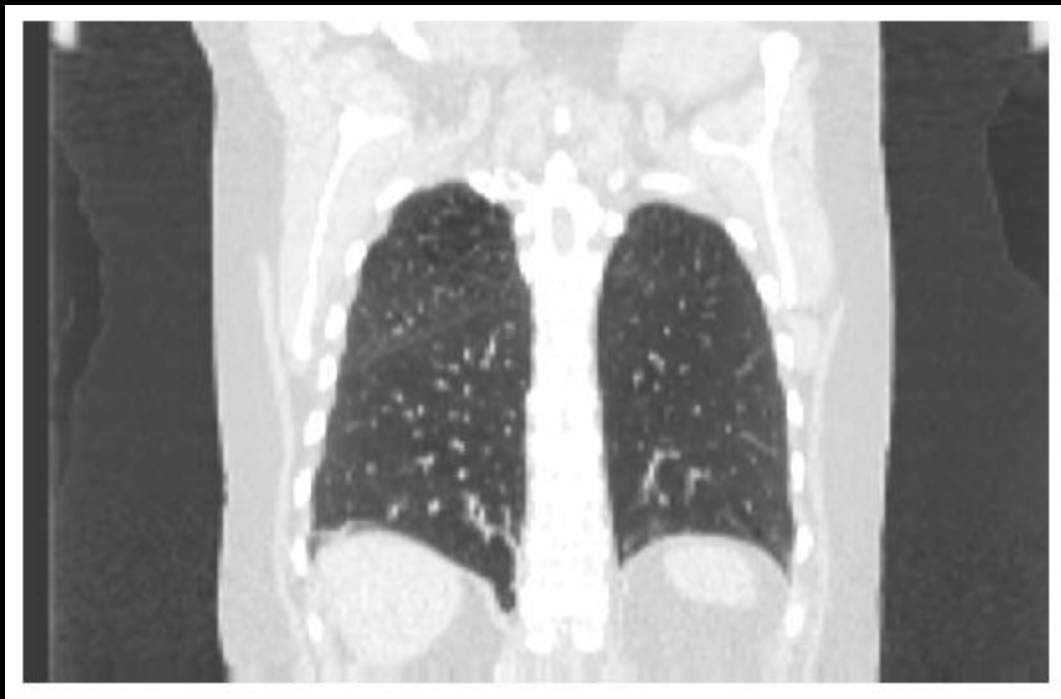
M

F



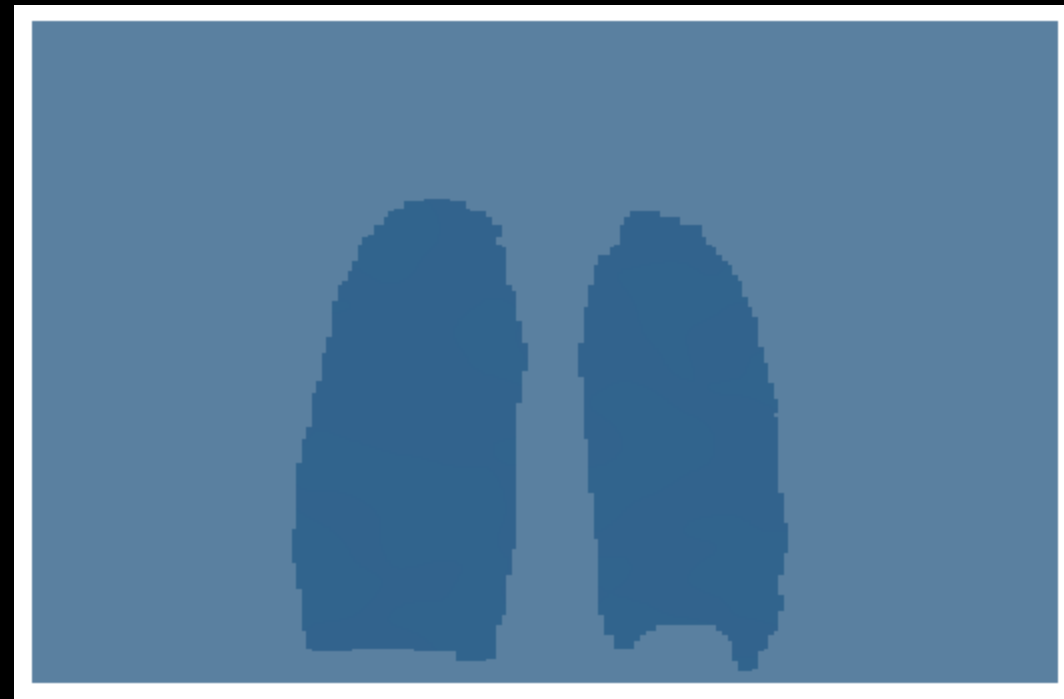
TRE = 1.11 mm

SIREN: $\omega = 64$



M

F



TRE = 9.89 mm

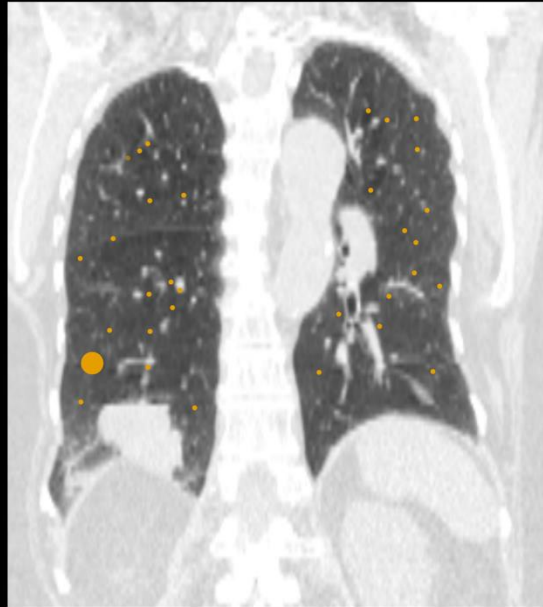
Regularization: analytical gradients

$$\mathcal{S}^{\text{hyper}}[\Phi] = \int_{\Omega} \left[\frac{1}{2} \alpha_1 |\nabla u|^2 + \alpha_a \phi_c(\text{cof } \nabla \Phi) + \alpha_v \psi(\det \nabla \Phi) \right] d\mathbf{x}$$

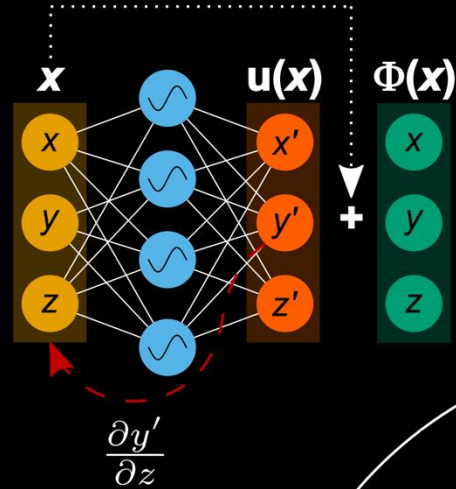
$$\mathcal{S}^{\text{jac}}[\Phi] = \int_{\Omega} |1 - \det \nabla \Phi| d\mathbf{x}$$

$$\mathcal{S}^{\text{bending}}[\Phi] = \frac{1}{8} \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 \left[\left(\frac{\partial^2 \Phi}{\partial x^2} \right)^2 + \left(\frac{\partial^2 \Phi}{\partial y^2} \right)^2 + \left(\frac{\partial^2 \Phi}{\partial z^2} \right)^2 + 2 \left(\frac{\partial^2 \Phi}{\partial xy} \right)^2 + 2 \left(\frac{\partial^2 \Phi}{\partial xz} \right)^2 + 2 \left(\frac{\partial^2 \Phi}{\partial yz} \right)^2 \right] dx dy dz$$

Fixed image F



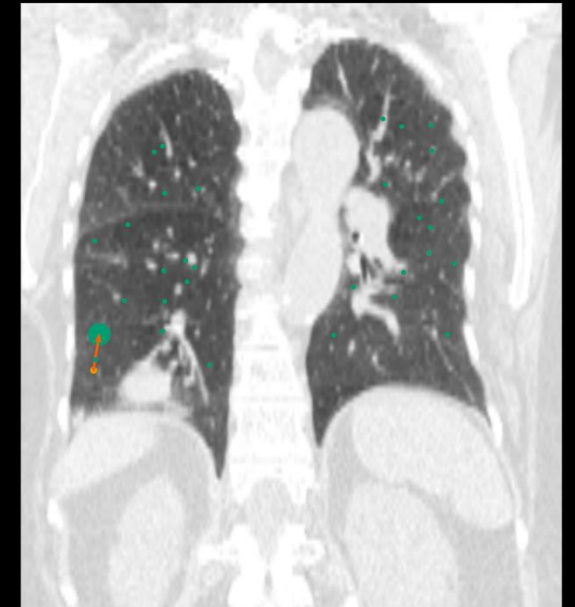
Transformation Φ



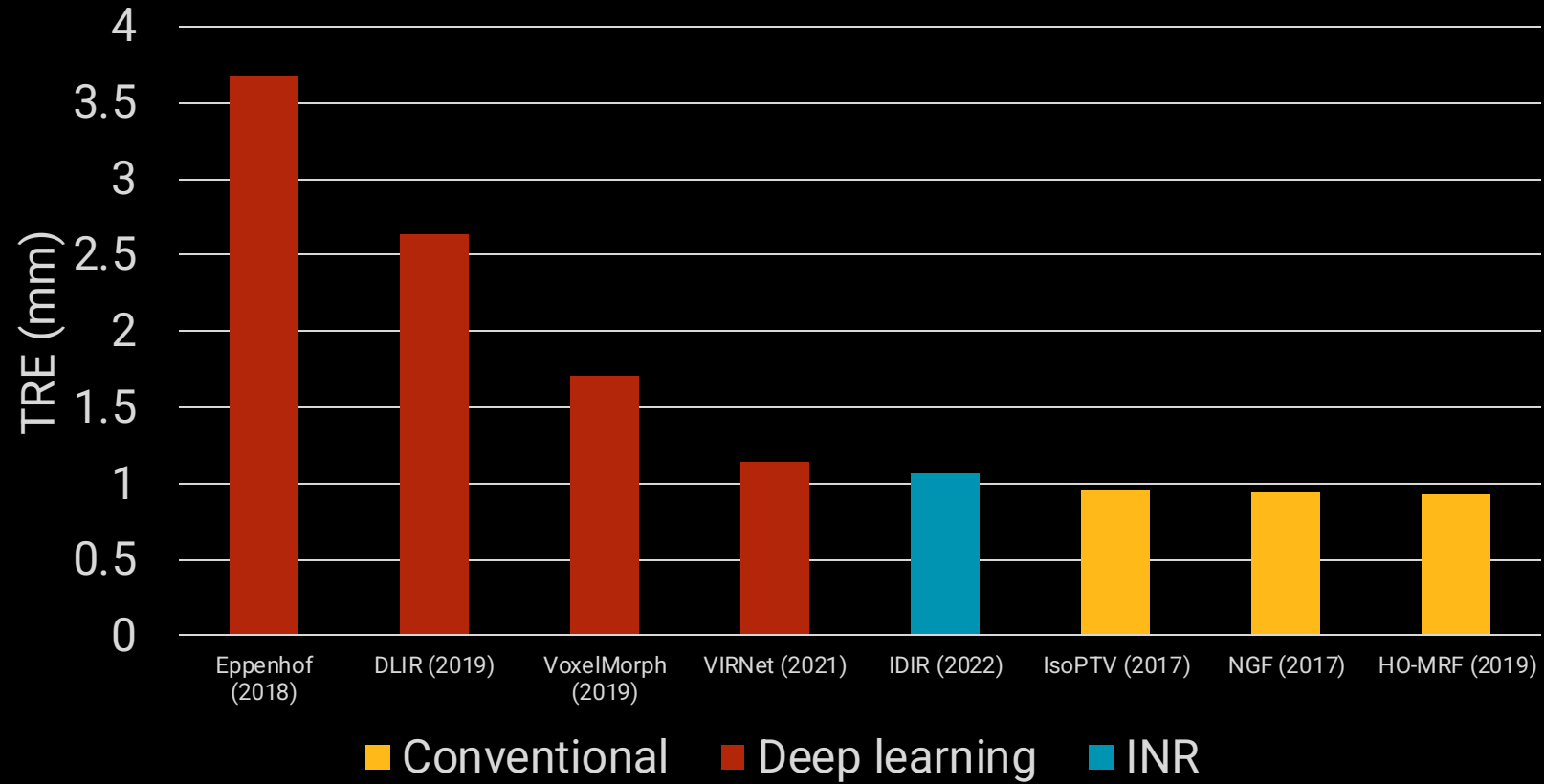
$$\hat{\Phi} = \underset{\Phi}{\operatorname{argmin}} \mathcal{L}_{\text{data}}(M \circ \Phi, F) + \alpha \mathcal{L}_{\text{reg}}(\Phi)$$

$$\det \nabla \Phi = \begin{vmatrix} \frac{\partial x'}{\partial x} & \frac{\partial y'}{\partial x} & \frac{\partial z'}{\partial x} \\ \frac{\partial x'}{\partial y} & \frac{\partial y'}{\partial y} & \frac{\partial z'}{\partial y} \\ \frac{\partial x'}{\partial z} & \frac{\partial y'}{\partial z} & \frac{\partial z'}{\partial z} \end{vmatrix}$$

Moving image M



Results



Histopathology registration

NEMESIS: NEural iMPLICIT rEpresentations for whole Slide Image regiStration



Bruno de Santi



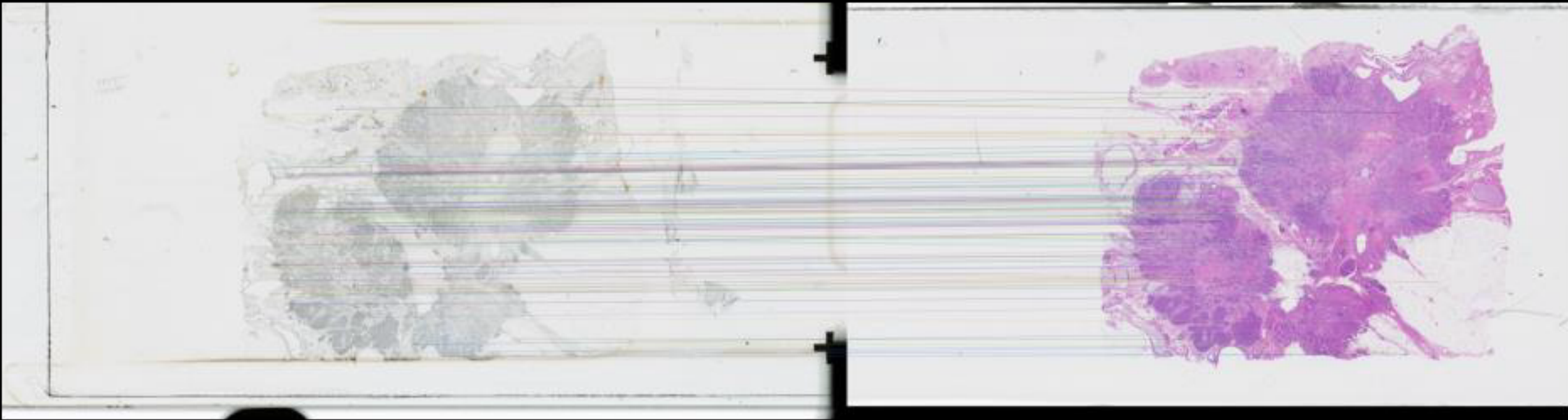
ACROBAT 2022
Automatic Registration Of Breast cAncer Tissue

❓ Use a minimal INR to first perform affine registration

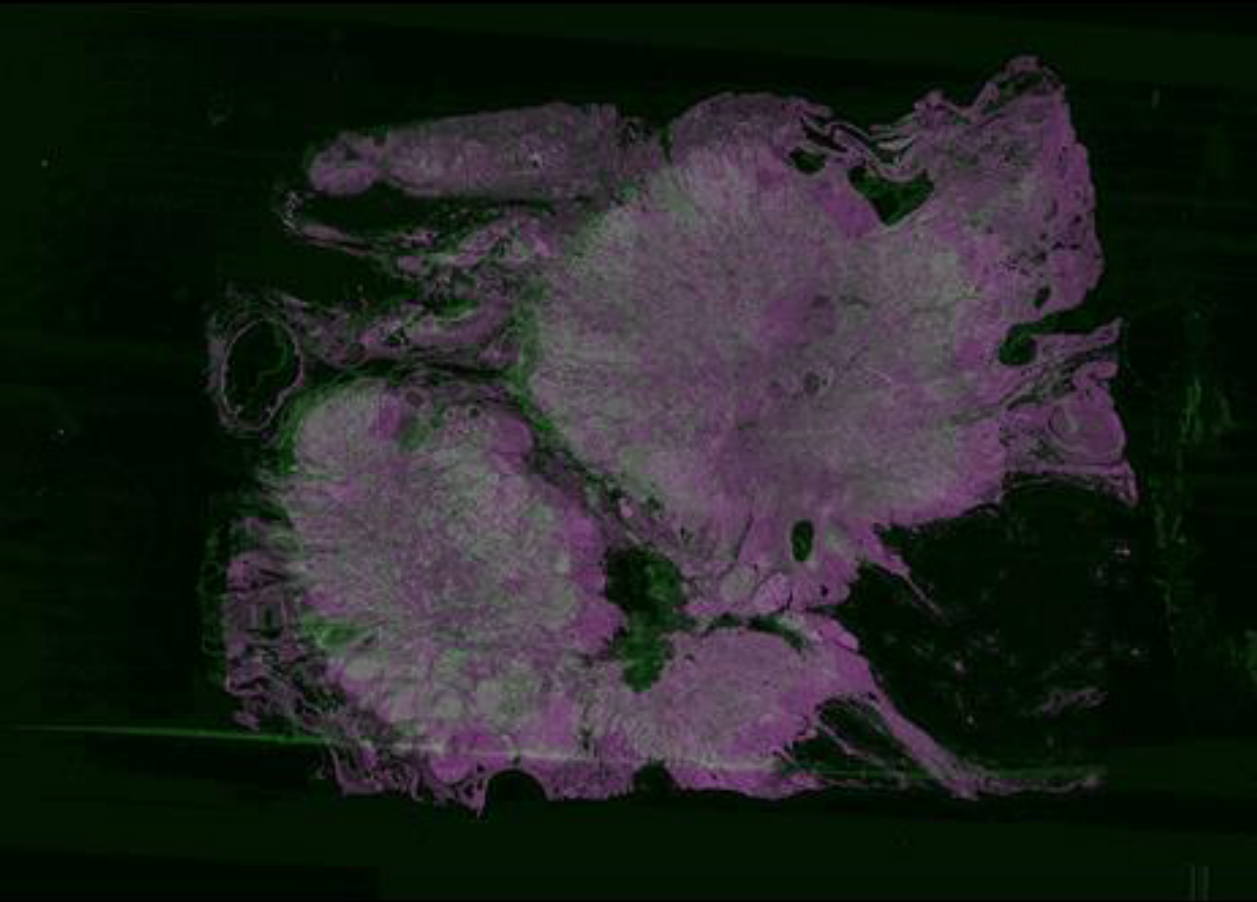
🤖 Lesson learned: some things are too hard for INRs

Step one: affine pre-alignment

SIFT features and RANSAC



Step two: INR fitting



- INR always improved results
- INR for huge images at original resolution
- MLP CT $\approx$ MLP histopathology

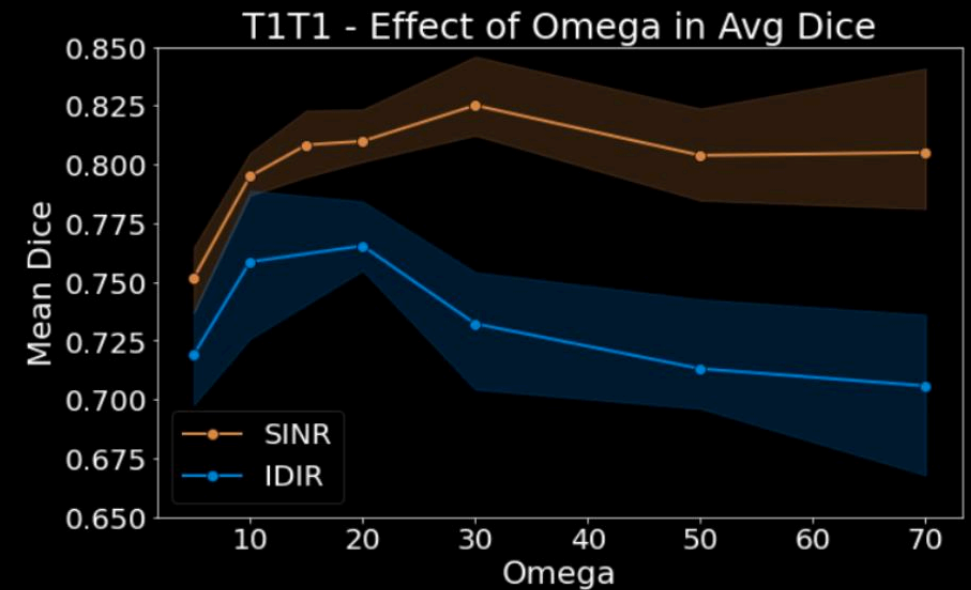


ACROBAT 2022
Automatic Registration Of Breast cAncer Tissue

Weitz, Philippe, et al. "The ACROBAT 2022 challenge: automatic registration of breast cancer tissue." Medical image analysis 97 (2024): 103257.

Ways to regularize your deformation

- Need smoothness? Exploit spectral bias
- Add explicit regularization terms
 - Jacobian determinant
 - Bending energy
- Select right ω for SIREN
 - High omega $\rightarrow$ More detail, noise, folding
 - Low omega $\rightarrow$ Less detail, smooth, no folding
- Use INR to parametrize deformation model¹
- Use other activation functions
 - E.g., growing cosine: $c(z) = z \cos(z)$

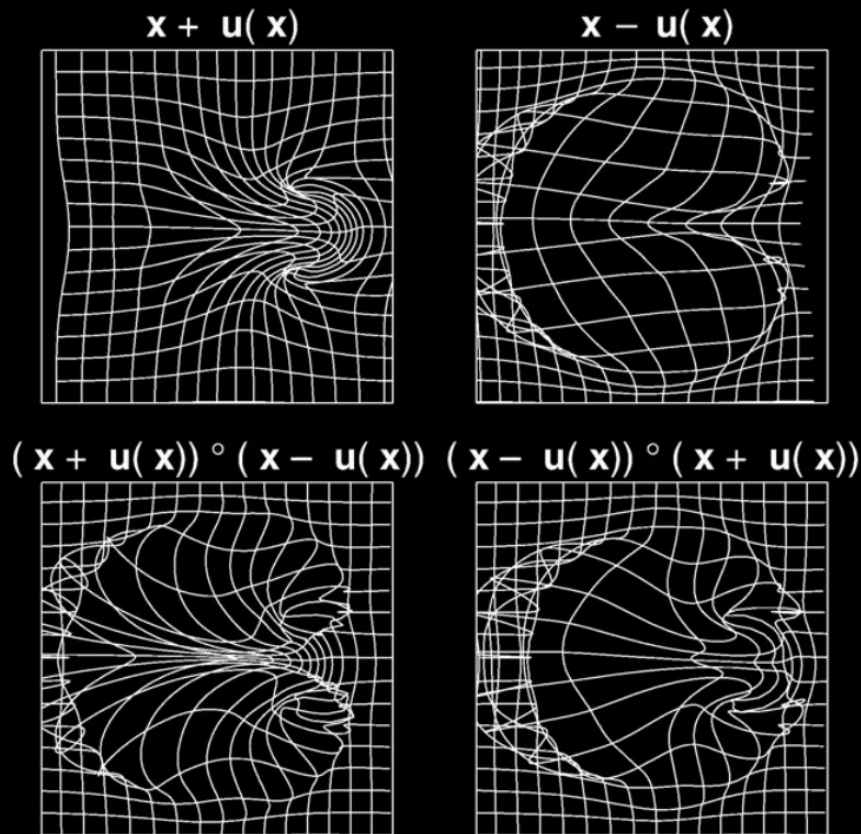


Sideri-Lampretsa, V., McGinnis, J., Qiu, H., Paschali, M., Simson, W., & Rueckert, D. (2024). SINR: Spline-enhanced implicit neural representation for multi-modal registration. In MIDL
Noel, M. M., Trivedi, A., & Dutta, P. (2021). Growing cosine unit: A novel oscillatory activation function that can speedup training and reduce parameters in convolutional neural networks. arXiv

Diffeomorphic registration

Small deformation model

$$\phi(x) = x + u(x)$$



Large deformation model

Time-dependent velocity

$$\frac{d\Phi}{dt} = u^{(t)}(\Phi^{(t)})$$

Diffeomorphism evolves between

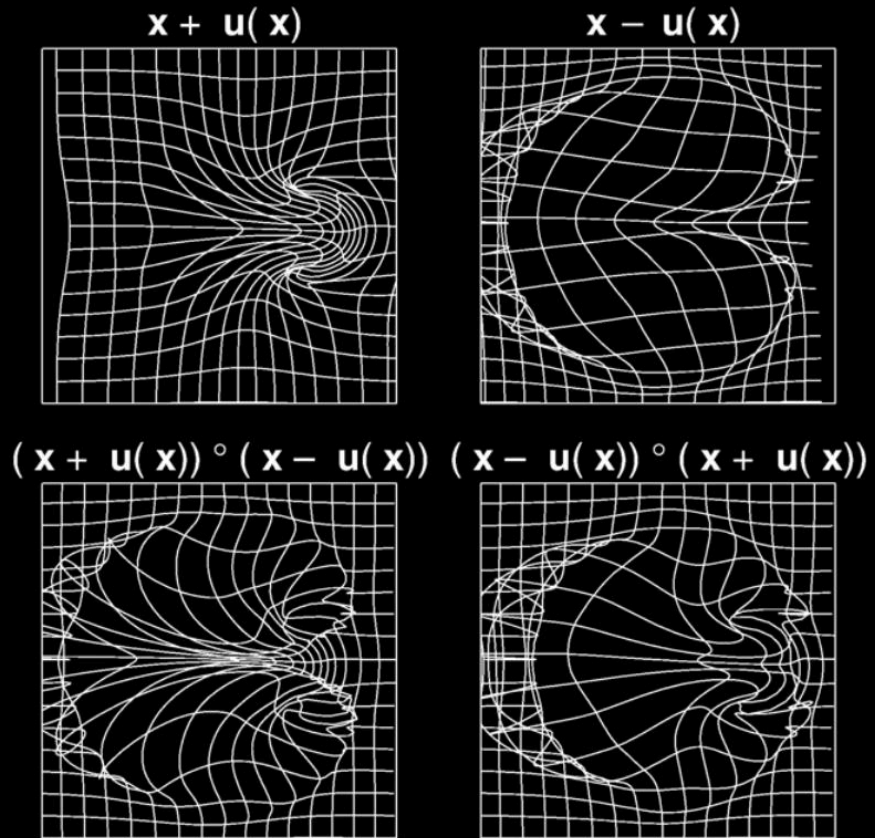
$$t = 0, \text{ i.e., } \phi^{(0)} = x$$

$$t = 1, \text{ i.e., } \phi^{(1)} = x + u(x)$$

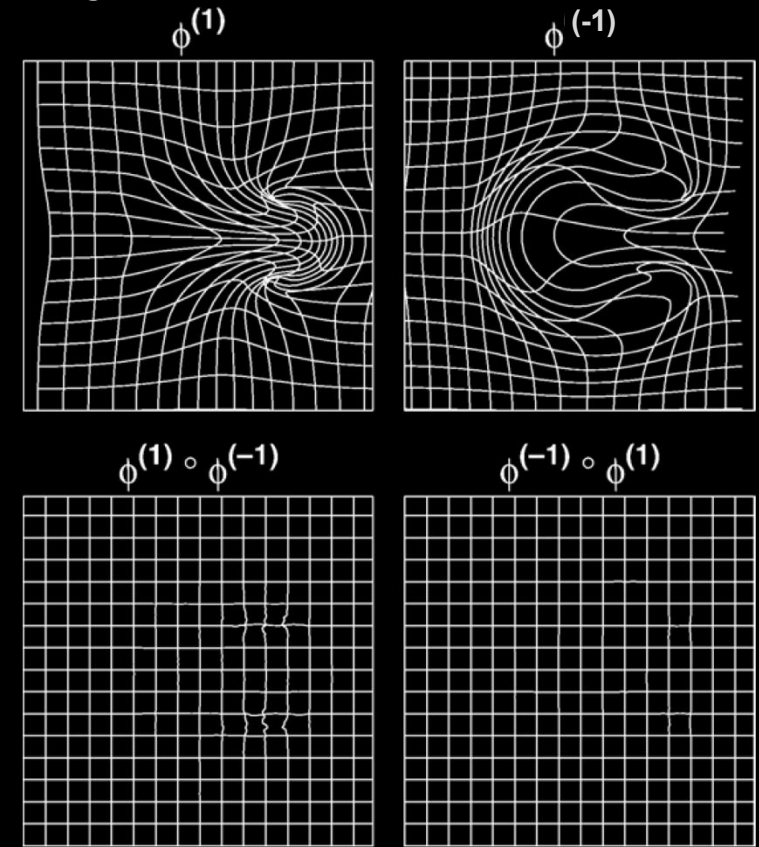
Invertible

Diffeomorphic registration

Small deformation model



Large deformation model

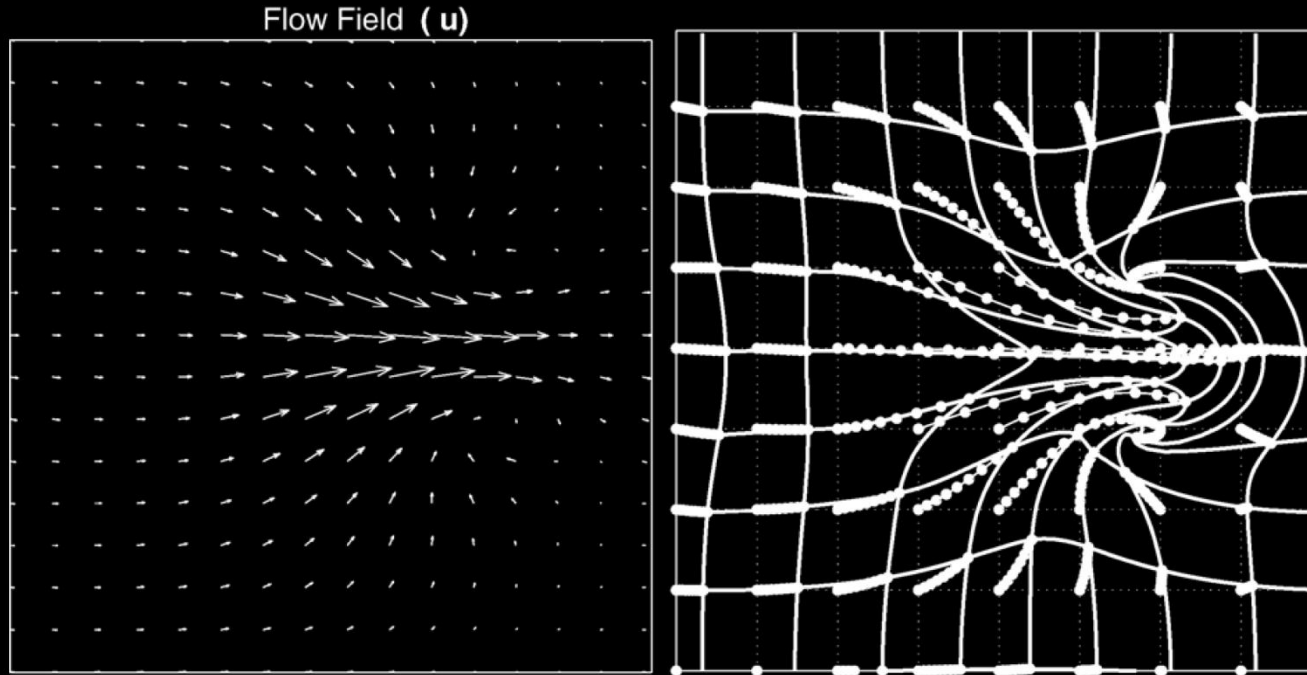


From velocity field to deformation field

$$\frac{d\Phi}{dt} = \mathbf{u}^{(t)}(\Phi^{(t)})$$

❓ Velocity field as INR

Stationary or time-dependent



Integration? Neural ODE solver!

```
from torchdiffeq import odeint
odeint(func, y0, t)
```

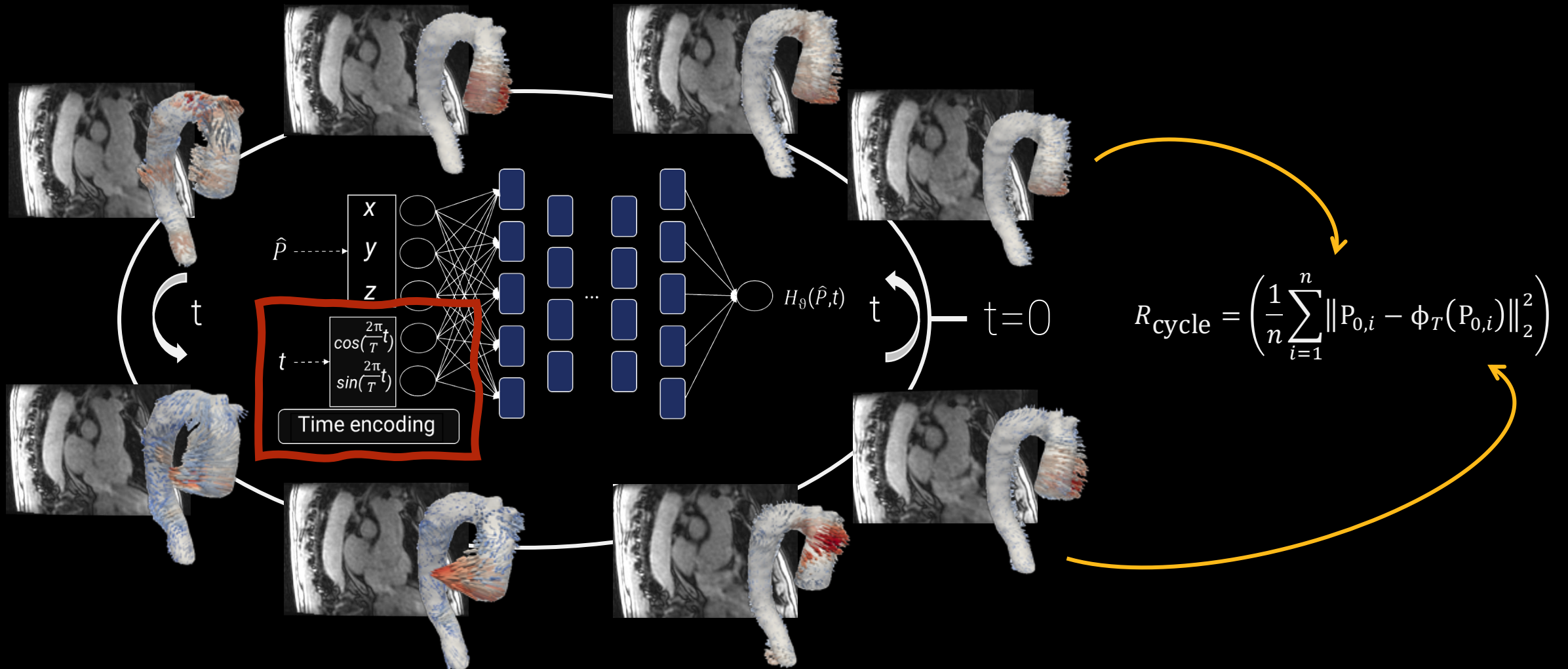
INR

$\phi^{(0)} = x$ $\{0, \dots, 1\}$

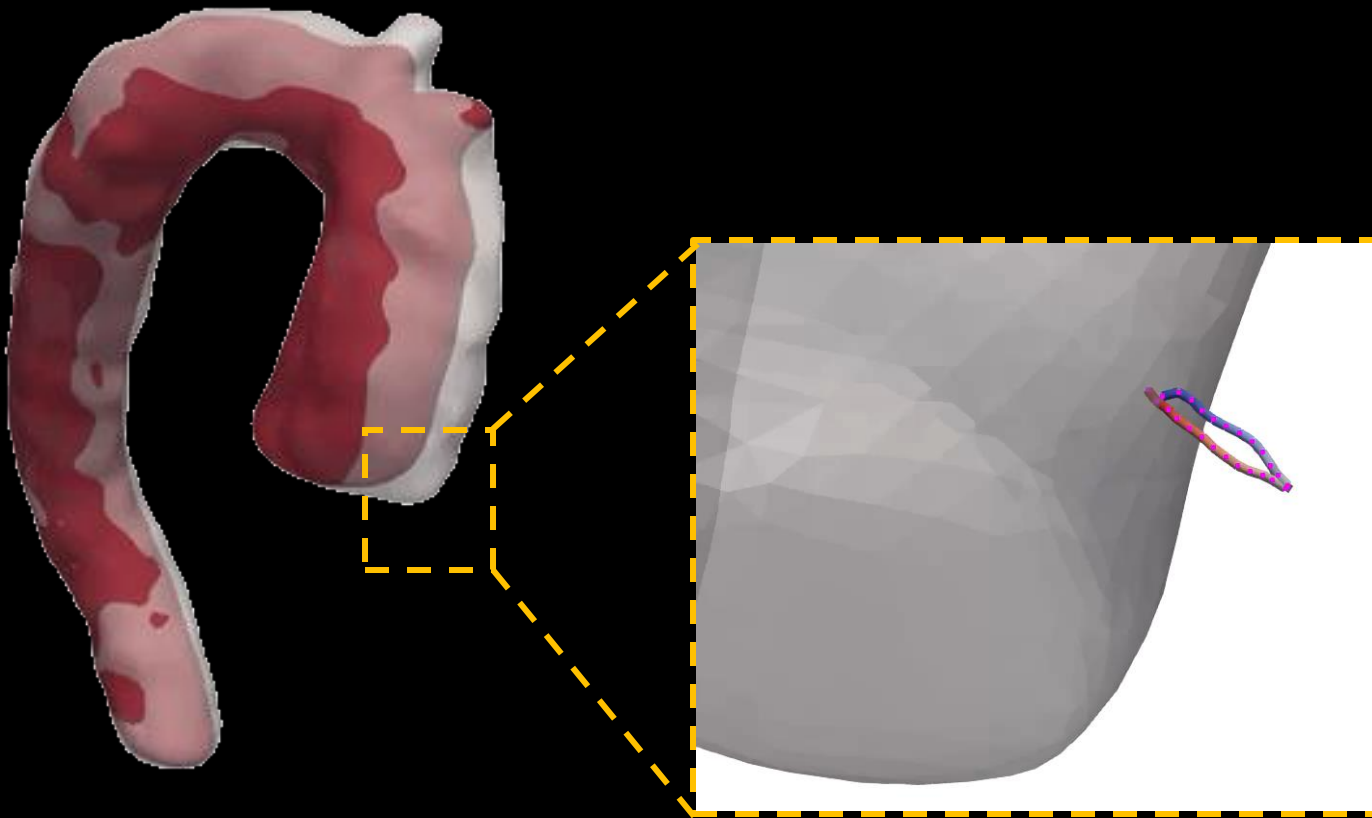
Or: squaring and scaling (requires grid)¹

¹ Han, Kun, et al. "Diffeomorphic image registration with neural velocity field." Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2023.

Application: 3D+t cardiac motion estimation



Periodic regularization



Simone Garzia



Garzia, S., Rygiel, P., Dummer, S., Cademartiri, F., Celi, S., & Wolterink, J. M. (2024). Neural Fields for Continuous Periodic Motion Estimation in 4D Cardiovascular Imaging. STACOM

Summary

Shape

- INRs can represent shape in 2D, 3D, 3D + t
- Shape-specific regularizers can improve your fitting
- Auto-decoder INR for generative shape model on same coordinate system
- Disentangle shape from translation and rotation (and scale?)

Registration

- Two (main) options
 1. An INR represents a (parametrized) *deformation* field
 2. An INR represents a (time-dependent) *velocity* field
- Use gradients to regularize deformation/velocity field
- Exploit 'conventional' wisdom (similarity metrics, optimizers, etc.)

Thanks

Dieuwertje Alblas

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Simone Garzia

Bruno de Santi

David Wiesner

Thank you for
your attention!

